



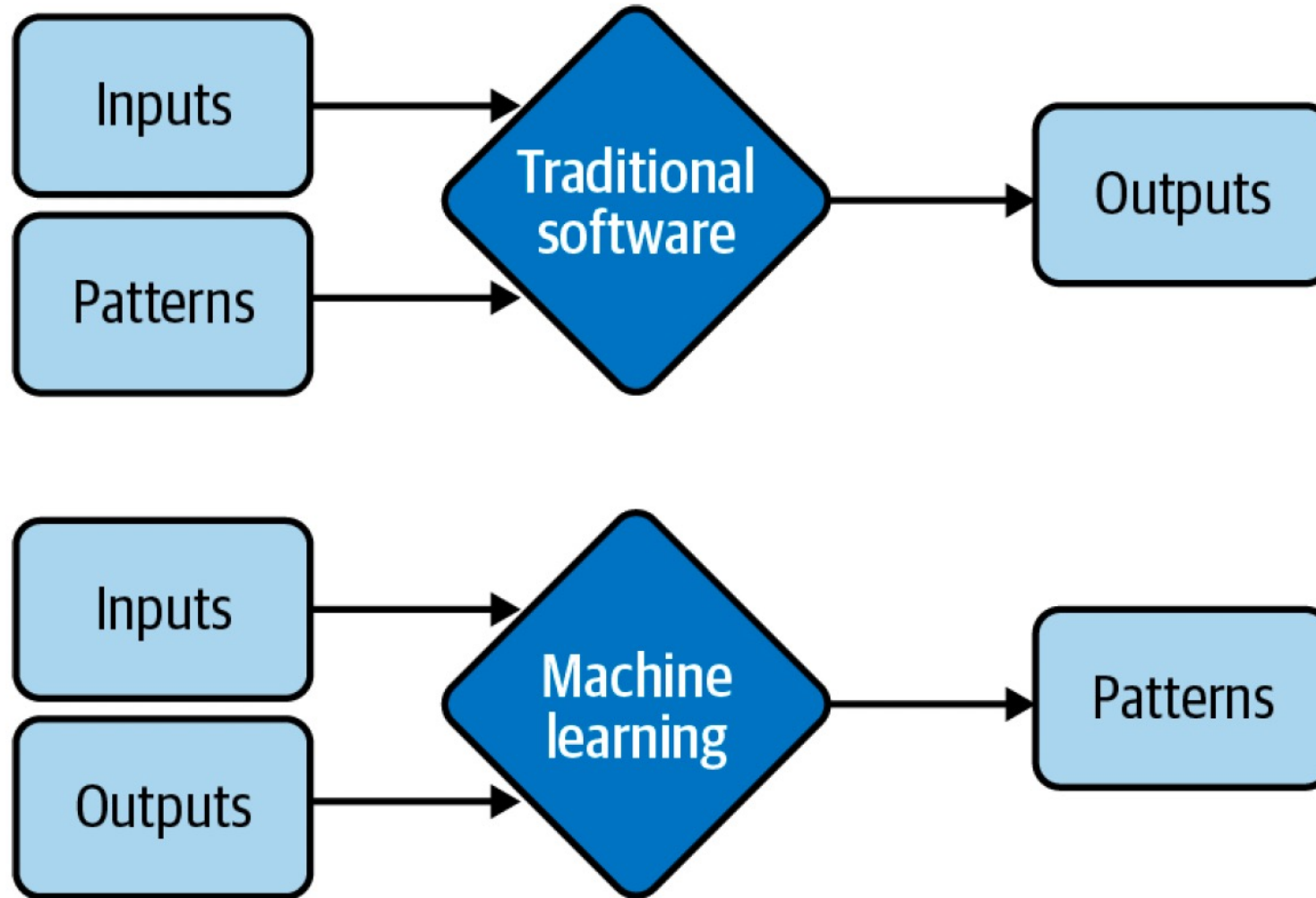
Exploratory Data Analysis

Basic engineering

Module 1

DS course 2023

Data is the foundation of Data Science



Exploratory Data Analysis

Exploratory Data Analysis is an approach to analyze the data and to summarize its main characteristics.

It refers to the critical process of using summary statistics and graphical representations for performing initial investigations on data:

- to discover patterns
- to spot anomalies
- to check assumptions

The term was first used by John Tukey in his book “Exploratory Data Analysis”, 1977.

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1 Data types

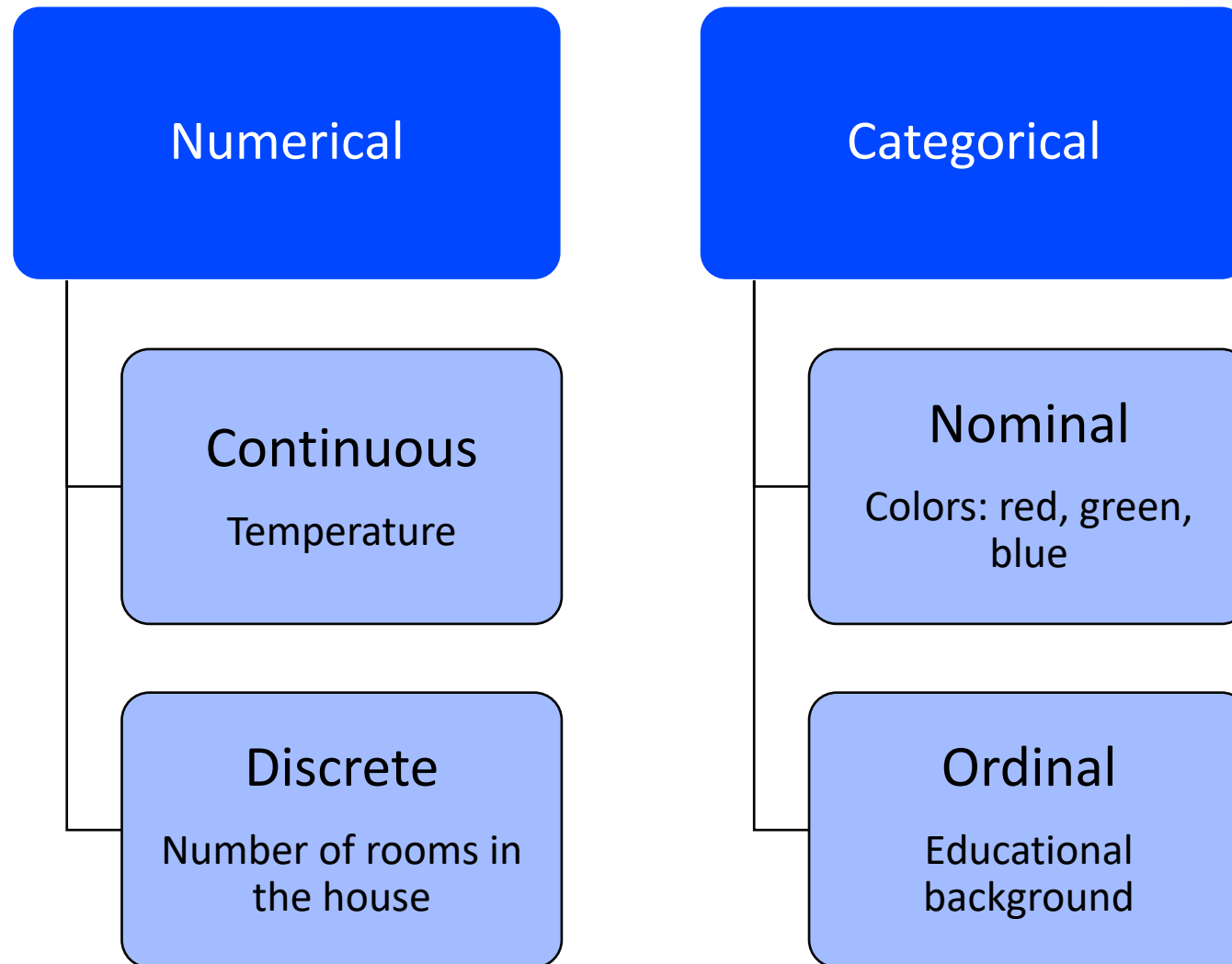
2 Descriptive statistics

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4 Data preparation

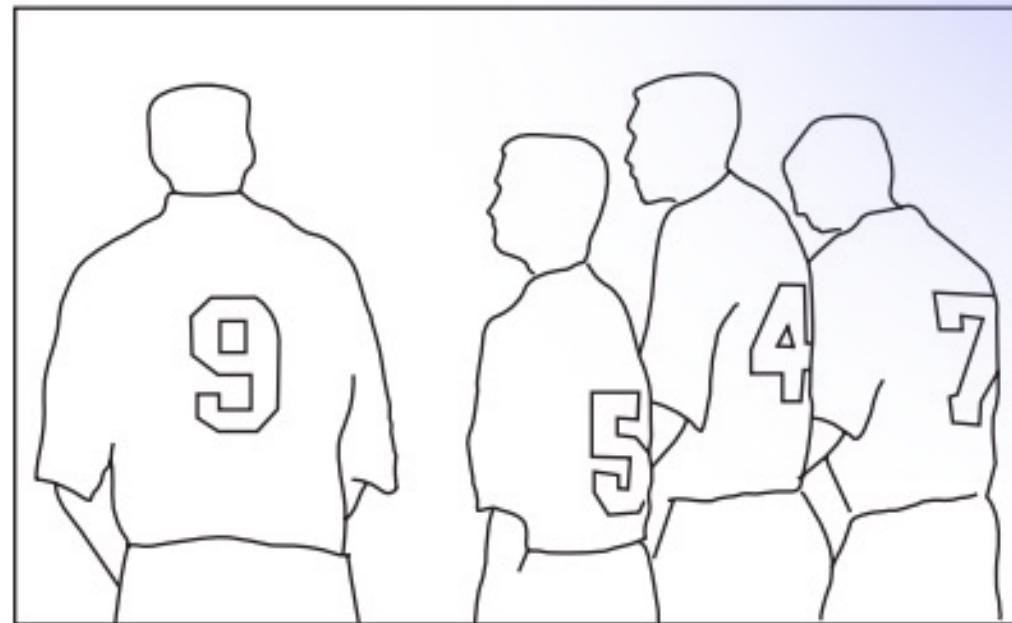
5 Dimensionality reduction

Types of data



Types of data

Age	Number_of_steps	Gender	Other_physical_act_level	Risk_of_disease
24	2518	Female	Medium	Yes
57	17122	Female	Low	Yes
45	6712	Female	High	Yes
42	6148	Male	Medium	No
63	9891	Male	High	No
24	1292	Female	High	Yes
43	16225	Male	Low	Yes
40	2183	Male	Low	Yes
18	12955	Male	Low	No
28	7825	Female	High	No



Rectangular (Tabular) Data

The typical frame of reference for an analysis in data science is a rectangular data object, like a spreadsheet or database table.

Rectangular data is essentially a two-dimensional matrix

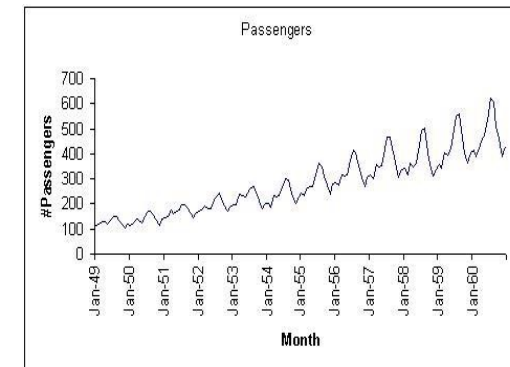
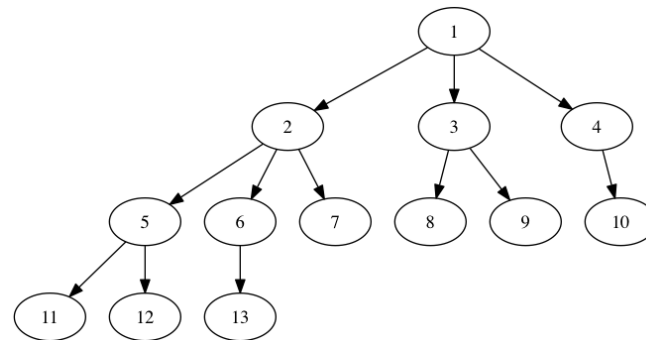
1.rows indicating records (case, example, instance, observation)

2.columns indicating features (variables, attributes)

Owner	Country	File_Date	IPC_Class
Company A	US	6/18/2008	H05H13
Company A	EP	1/30/1998	A61N5
Company A	EP	1/30/1998	A61N5
Company A	EP	1/30/1998	A61N5
Company A	JP	8/28/1997	A61N5
Company A	JP	10/4/2002	A61N5
Company A	JP	1/27/2003	A61N5
Company A	JP	4/14/2003	A61N5
Company A	JP	5/13/2011	A61N5
Company B	JP	4/2/1998	G12B13
Company B	JP	4/2/1998	G12B13
Company B	JP	5/28/1997	A61N5
Company B	JP	11/12/1997	A61N5
Company B	JP	2/29/2000	A61N5
Company B	JP	4/30/2002	A61N5

Examples of nonrectangular data:

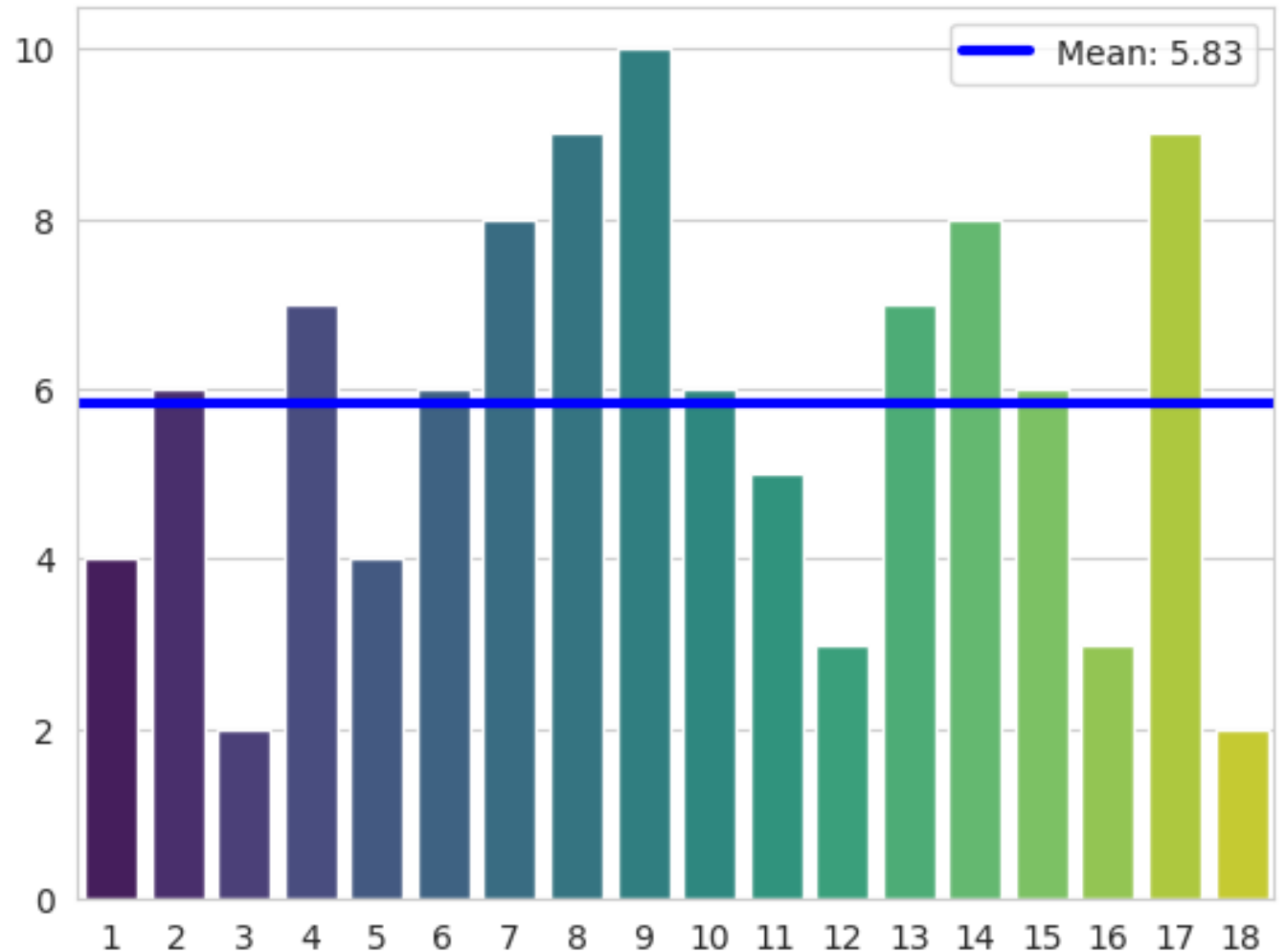
- Graph data
- Hierarchical data
- Spatial data
- Time series data



Descriptive Statistics: Measures of central tendency

$$\text{Mean} = \bar{x} = \frac{\sum_i^n x_i}{n}$$

data = [4, 6, 2, 7, 4, 6, 8, 9, 10, 6, 5, 3, 7, 8, 6, 3, 9, 2]



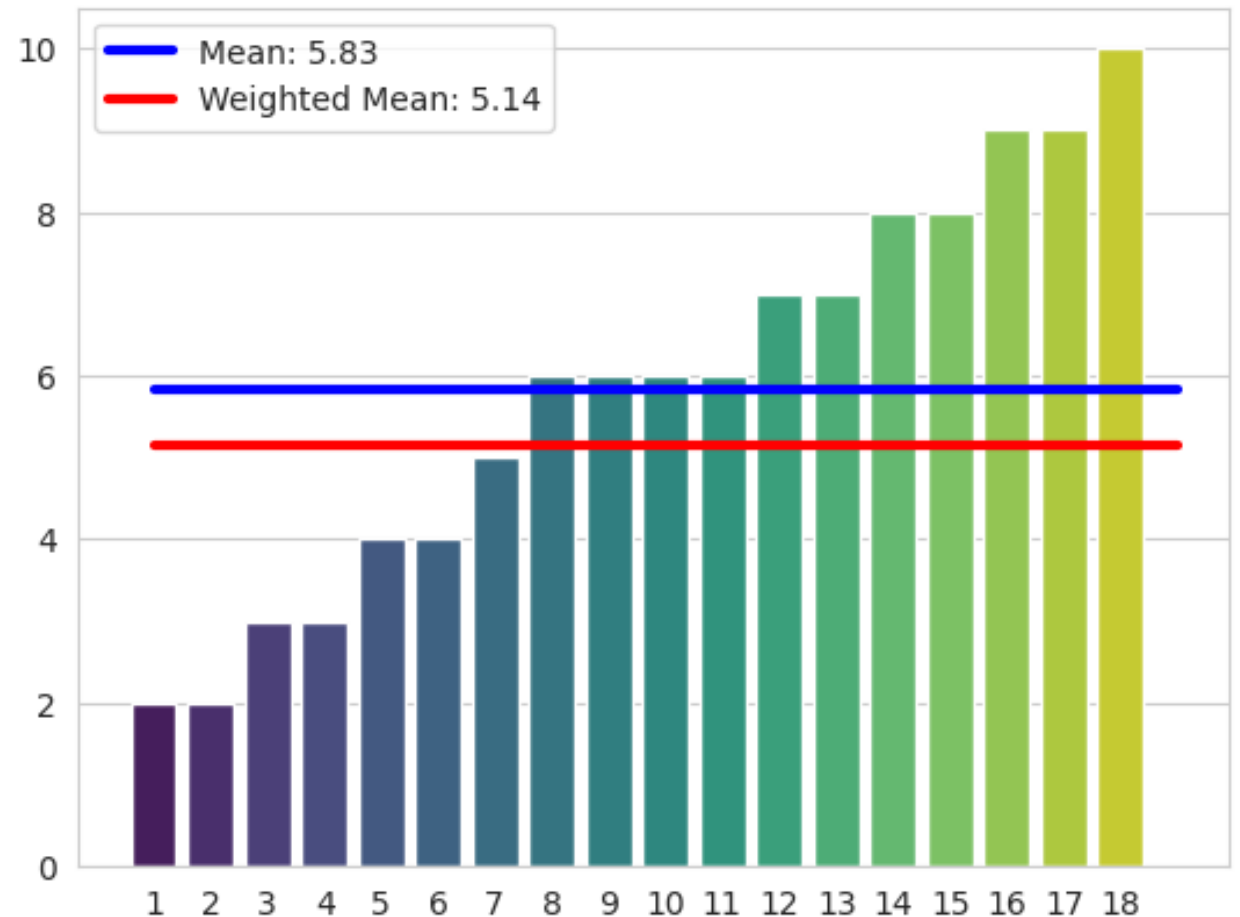
Descriptive Statistics: Measures of central tendency

Weighted mean

$$\bar{x}_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_i w_i}$$

Sorted data: [2, 2, 3, 3, 4, 4, 5, 6, 6, 6, 6, 7, 7, 8, 8, 9, 9, 10]

Weights: [3, 3, 3, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]



Descriptive Statistics: Measures of central tendency

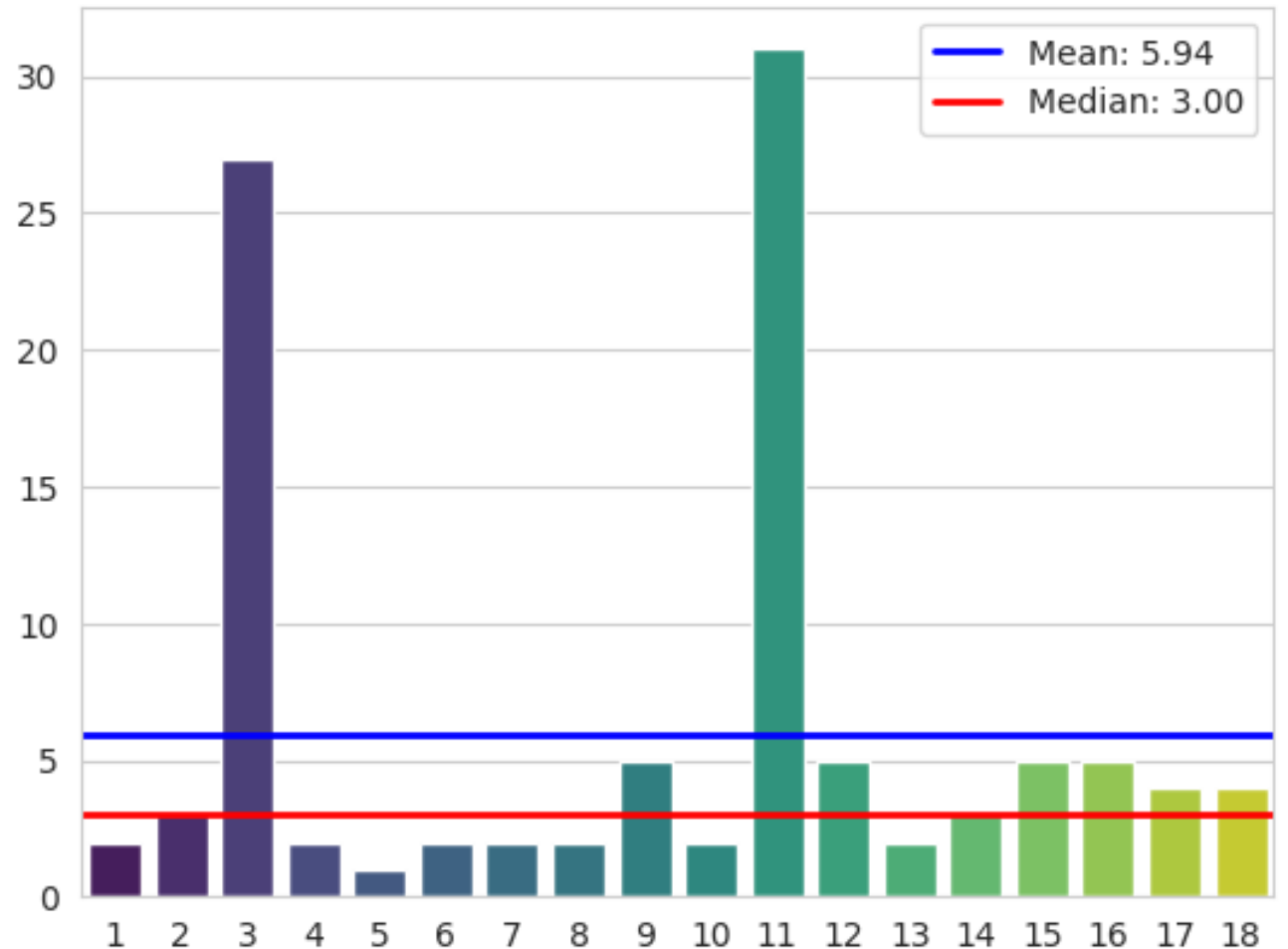
Median, the robust estimate

The median represents the middle value in a sorted dataset

Unlike the mean, the median is not skewed by extreme values (outliers), therefore it is often a better measure of central tendency for skewed datasets.

sorted_data = [1, 2, 2, 2, 2, 2, 2, 2, 3, 3, 4, 4, 5, 5, 5, 5, 27, 31]

data = [2, 3, 27, 2, 1, 2, 2, 2, 5, 2, 31, 5, 2, 3, 5, 5, 4, 4]



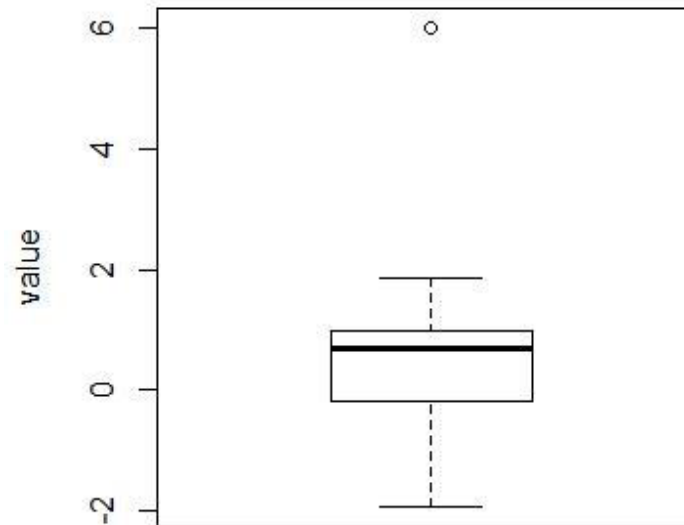
Outliers

An outlier is a data point that is significantly different from other points.

An outlier cannot be covered by data general pattern and/or distribution of the data.

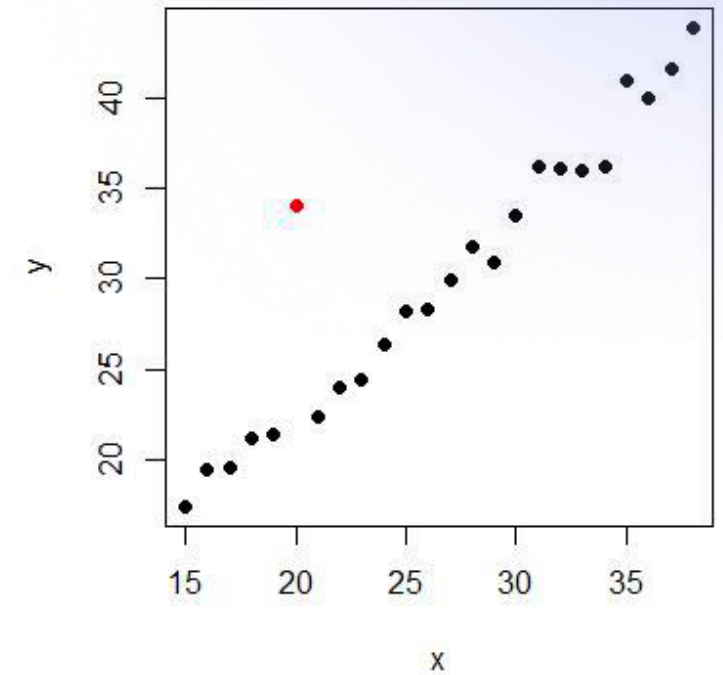
Outliers can arise due to variability in the data or due to measurement errors.

A. Boxplot



A

B. Scatter plot



Descriptive Statistics: Measures of central tendency

Mode

The mode is the value (or values),
that appear most often in the data



Descriptive Statistics: Estimates of Variability

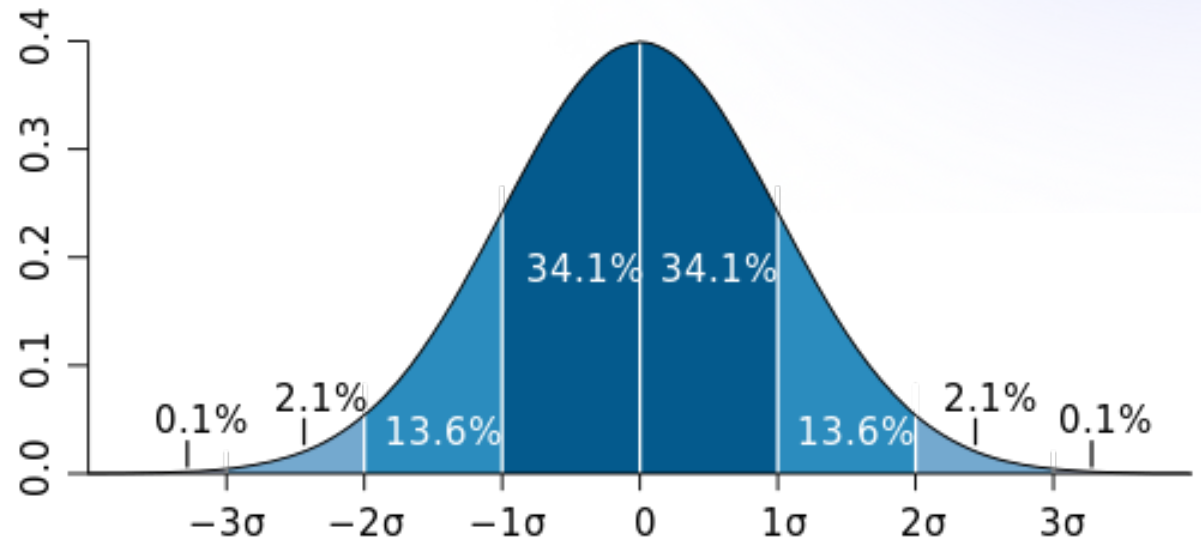
$$\text{Variance} = s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\text{Standard deviation} = s = \sqrt{\text{Variance}}$$

Measures whether the data values are tightly clustered or spread out around the mean

Based on the differences, or deviations, between the mean and the observed data

The standard deviation is easier to interpret than the variance, as it is on the same scale as the original data



Anscombe's Quartet

1973 Francis Anscombe created 4 datasets of 2d points.

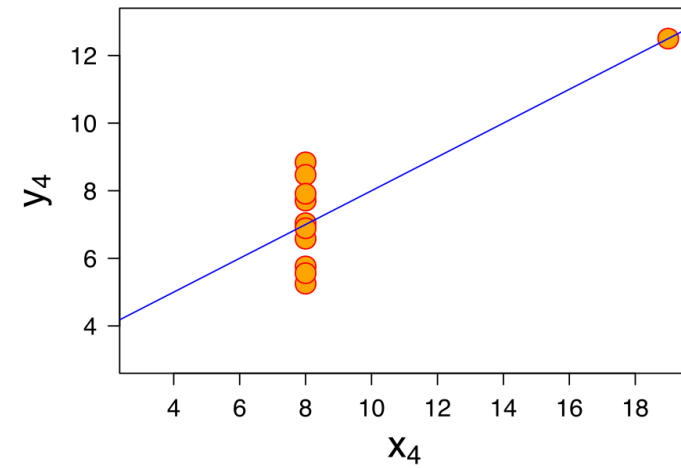
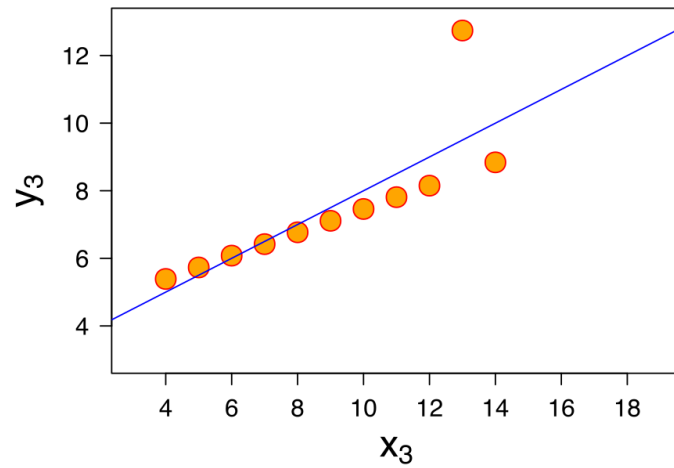
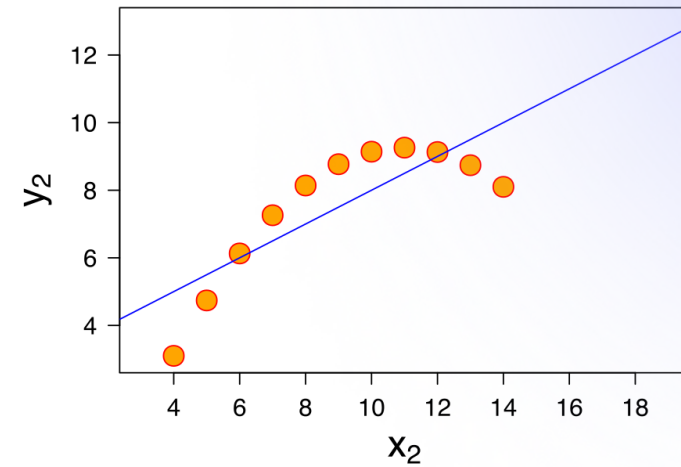
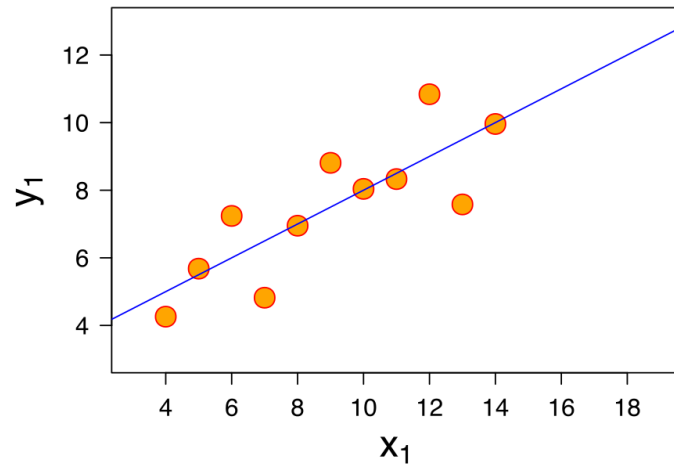
He demonstrated:

- the importance of graphing data when analyzing it
- the effect of outliers and other influential observations on statistical properties.

Set 1		Set 2		Set 3		Set 4	
x	y	x	y	x	y	x	y
4	4.26	4	3.1	4	5.39	8	6.58
5	5.68	5	4.74	5	5.73	8	5.76
6	7.24	6	6.13	6	6.08	8	7.71
7	4.82	7	7.26	7	6.42	8	8.84
8	6.95	8	8.14	8	6.77	8	8.47
9	8.81	9	8.77	9	7.11	8	7.04
10	8.04	10	9.14	10	7.46	8	5.25
11	8.33	11	9.26	11	7.81	8	5.56
12	10.84	12	9.13	12	8.15	8	7.91
13	7.58	13	8.74	13	12.74	8	6.89
14	9.96	14	8.1	14	8.84	19	12.5

Property	Value
Mean of x	9
Sample variance of x: s_x^2	11
Mean of y	7.50
Sample variance of y: s_y^2	4.125
Correlation between x and y	0.816
Linear regression line	$y = 3.00 + 0.500x$
Coefficient of determination of the linear regression: R^2	0.67

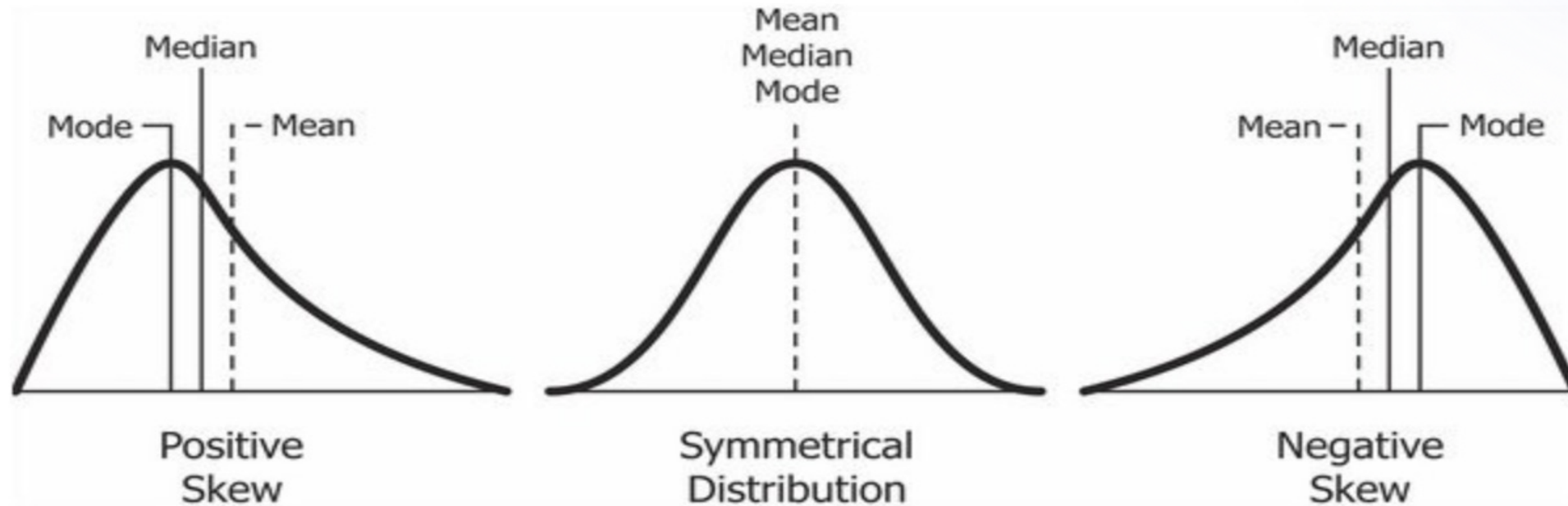
Anscombe's Quartet visualization



Distribution, Skewness and Kurtosis

Skewness (symmetry: positive or negative)

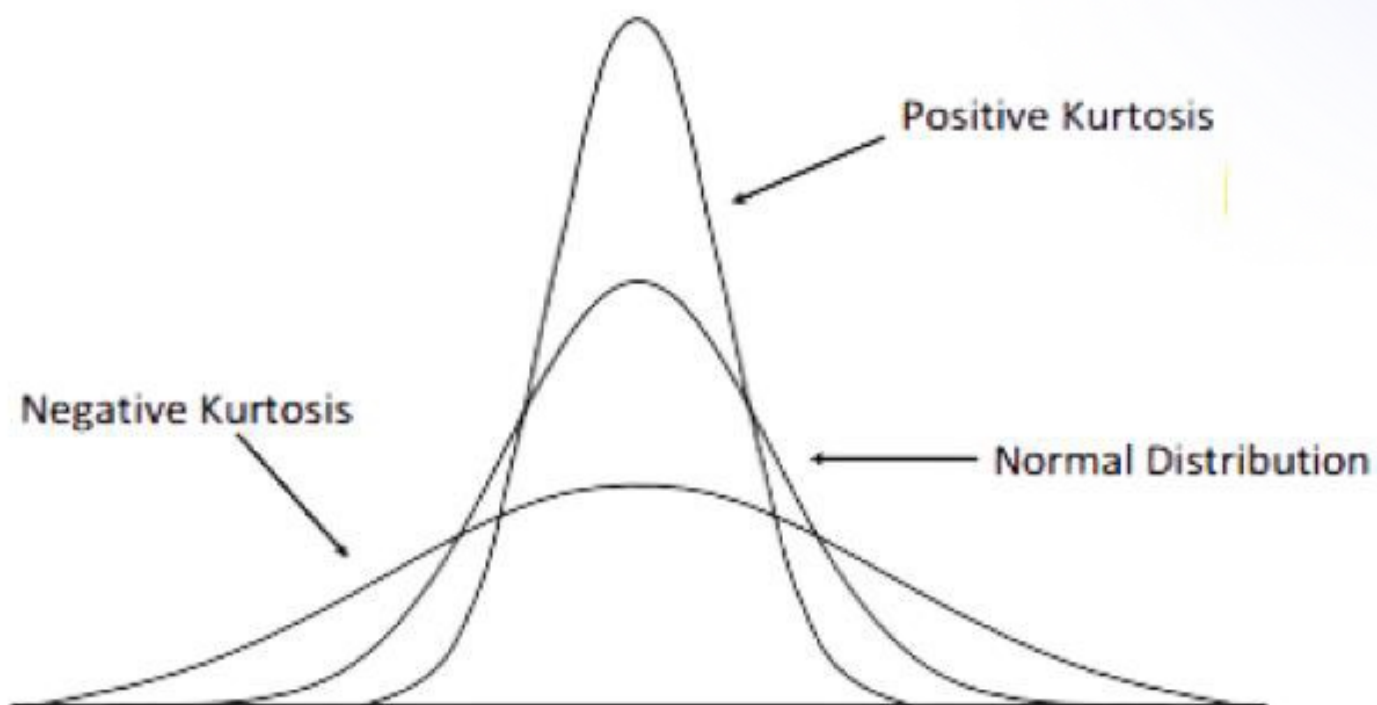
$$Skewness = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$



Skewness and Kurtosis

Kurtosis (shape: tall or flat)

$$Kurtosis = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - 3$$



Order Estimates

Order statistics: statistics based on sorted (ranked) data

Range: span between the largest and smallest number

Range = Max – Min

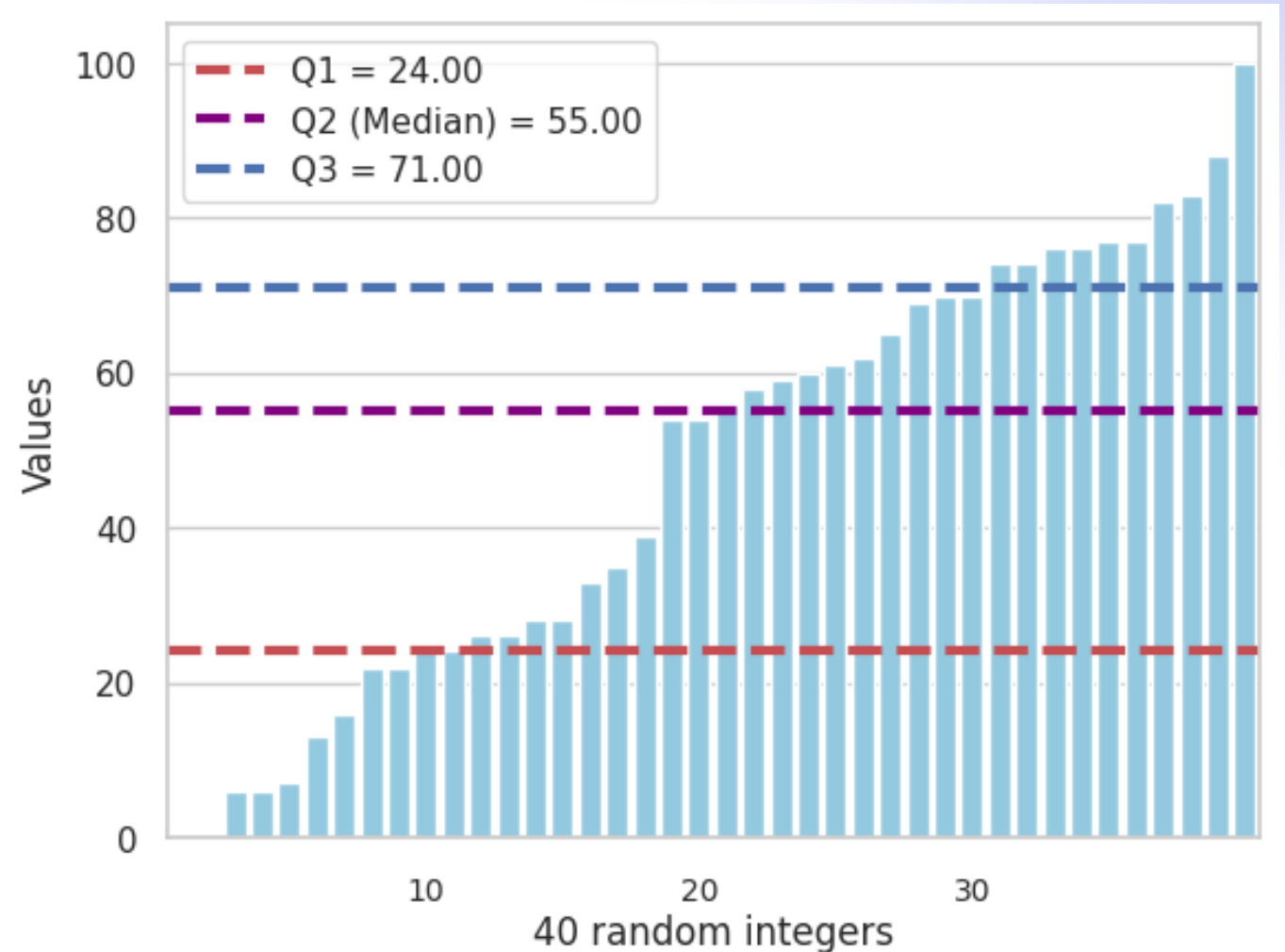
p-th percentile: percent of the values take on this value or less.

Median=Q2 is the 50th percentile,

Q1 is 25th

Q3 is 75th

Interquartile range (IQR) = Q3 – Q1



Data visualization

Univariate analysis

- Single feature (variable) analysis

Bivariate analysis

- relationship between 2 features

Multivariate analysis

- relationship between >2 features

Types of plots

- Box Plots
- Histograms
- Bar Plots
- Violin Plots
- Pie Charts
- Scatter Plots
- Pair Plots
- Heatmaps
- and others

Univariate analysis

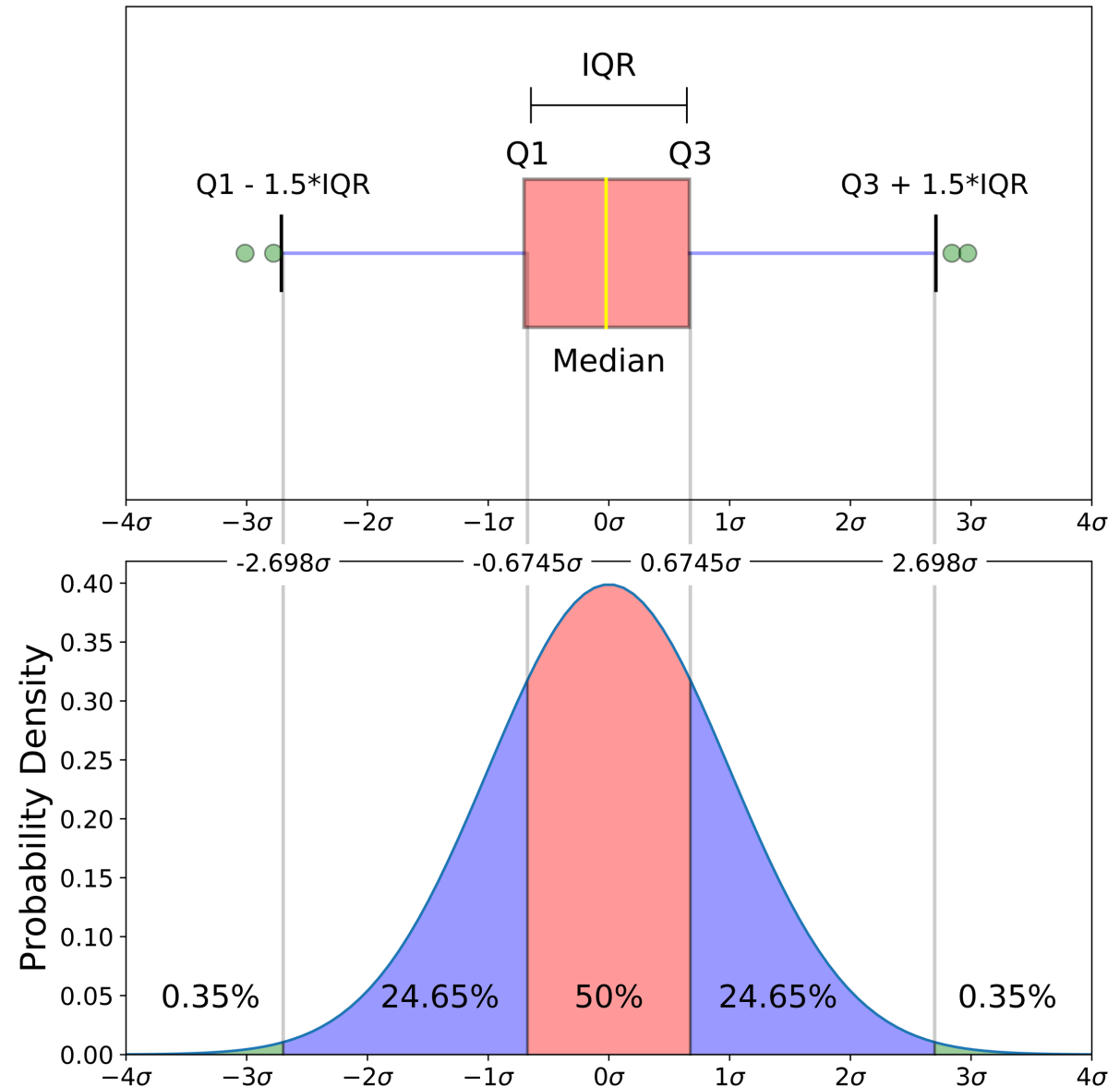
Numeric data

Boxplot

Boxplots (Tukey, 1977) are based on percentiles - quick way to visualize the distribution of data.

- The boundaries of the box are the 75th and 25th percentiles, respectively
- The median is shown by the horizontal line in the box
- The lines, referred to as whiskers, extend from both sides to indicate the range for the bulk of the data
- Any data outside of the whiskers is plotted as single points.

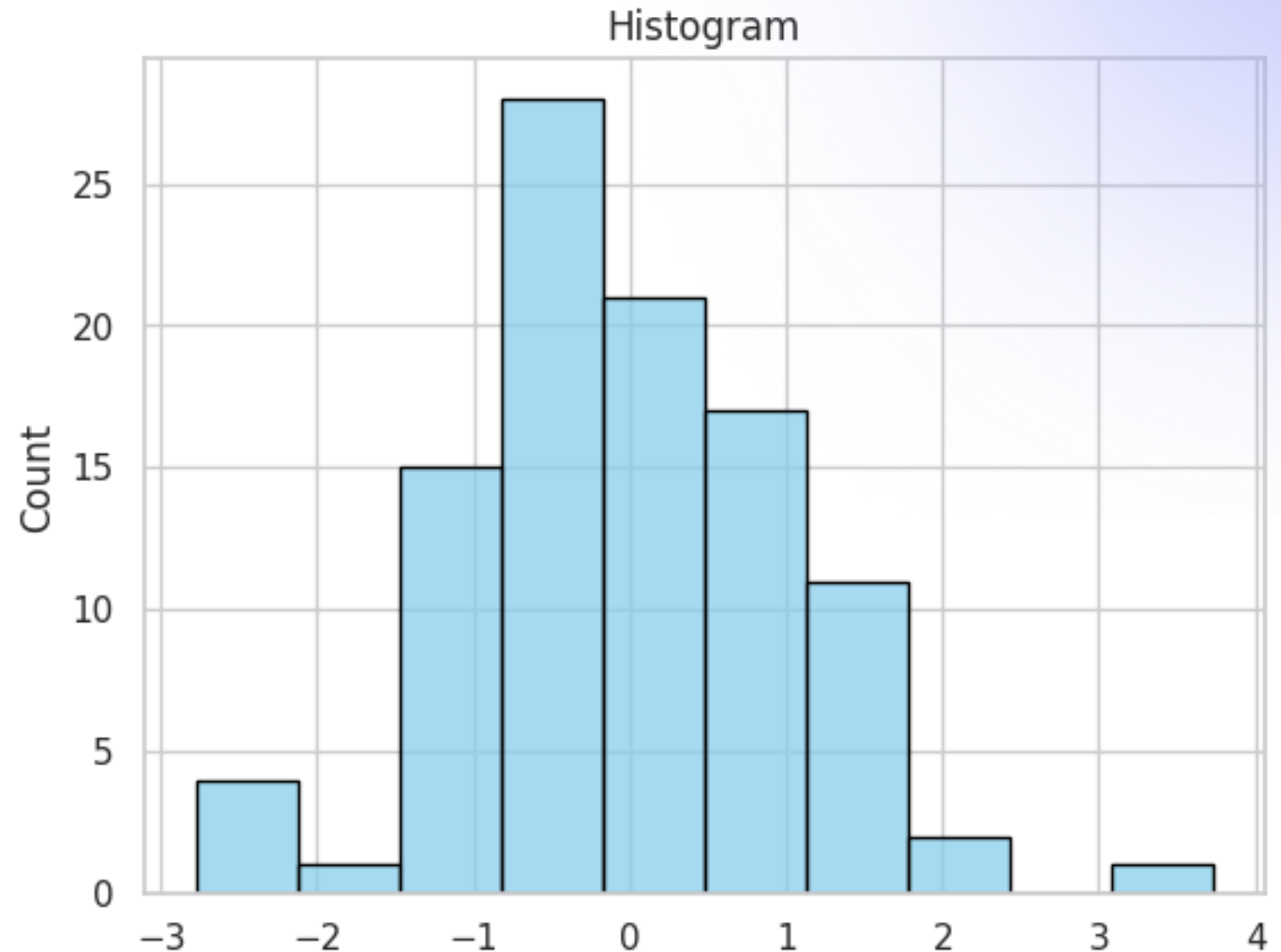
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Histograms

A **histogram** is an approximate representation of the distribution of numerical data.

- “Bin” the range of variable (divide up into equally spaced segments)
- count how many values fall into each interval
- Empty bins are included in the graph
- Number of bins is up to the user
- Bars are contiguous - no empty space shows between bars, unless there is an empty bin

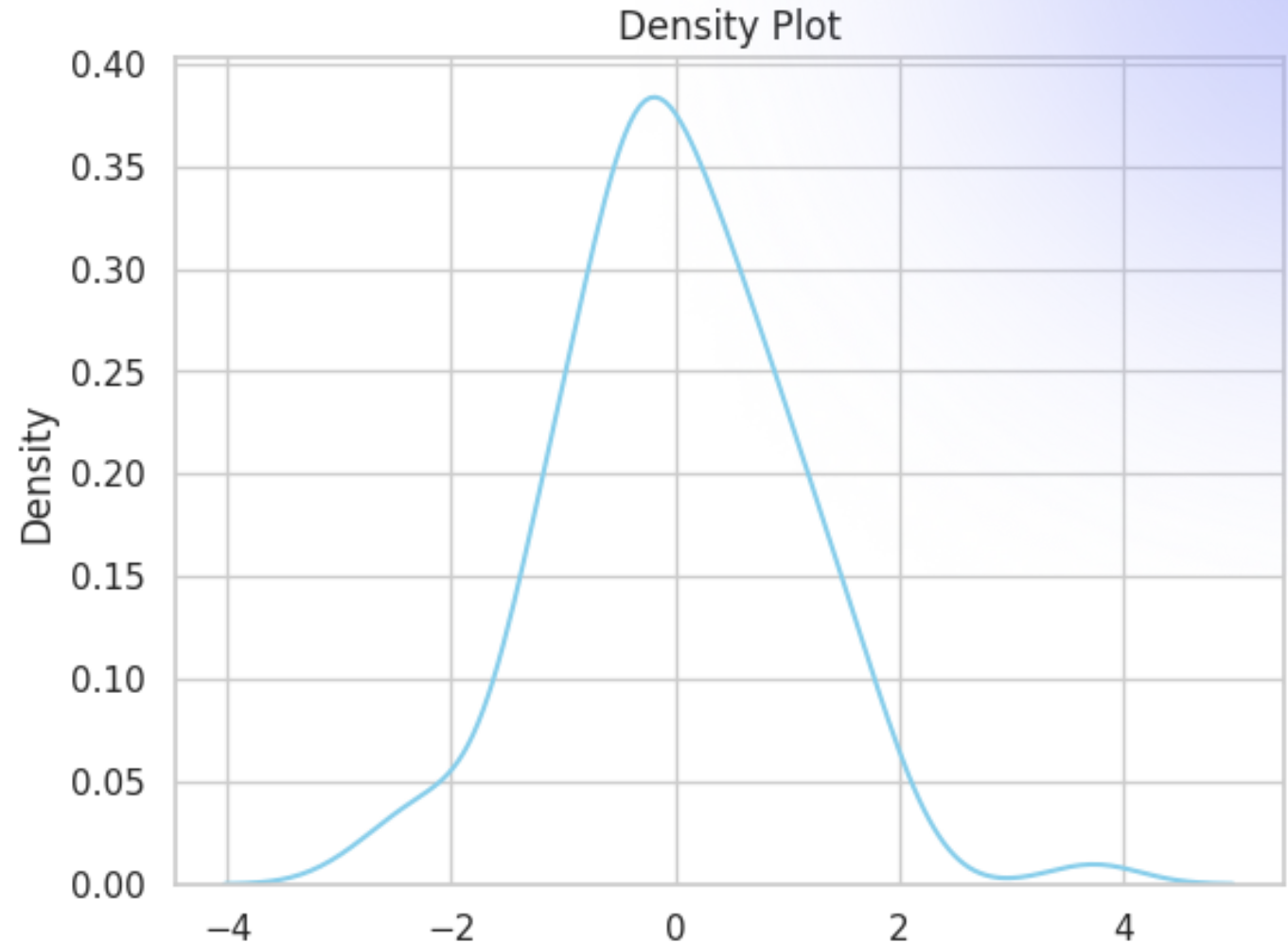


Density plot

Density plot, shows the distribution of data values as a continuous line

A smoothed histogram

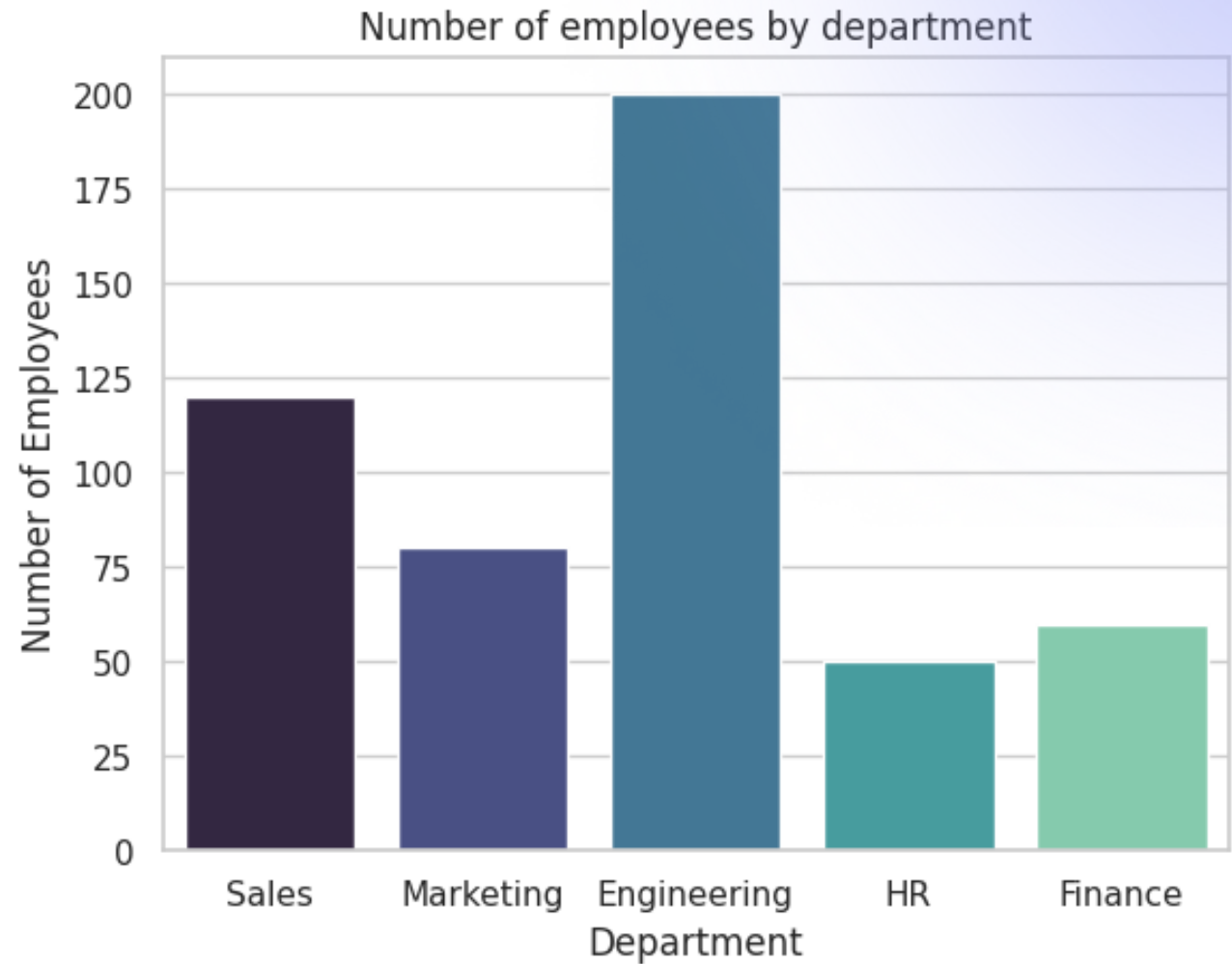
A key distinction from the histogram is the scale of the y-axis



Categorical Data

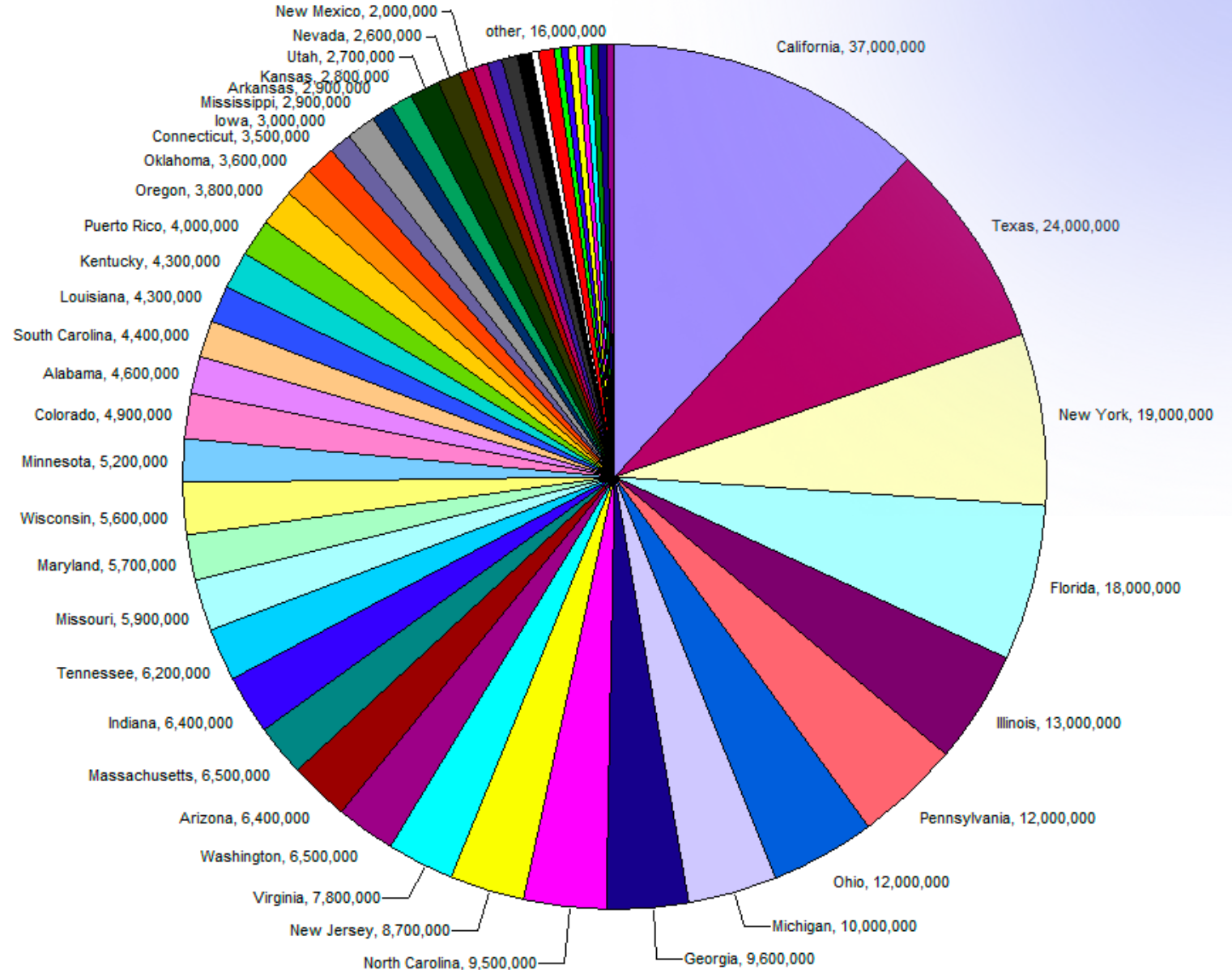
Bar Charts

- Represents categorical data with rectangular bars
- each bar corresponds to a category, and the size of the bar represents the value or frequency of that category
- the bars are shown separate from one another



Pie chart

- Represents categorical data as "slices" of a "pie". Whole pie represents total data, and each slice corresponds to a category.
- Size of each slice (or its central angle) is proportional to the category's ratio of the total.
- Primarily used to represent proportions or percentages.
- Best used when the number of categories is relatively small.
- Gives a quick visual impression of relative proportions.

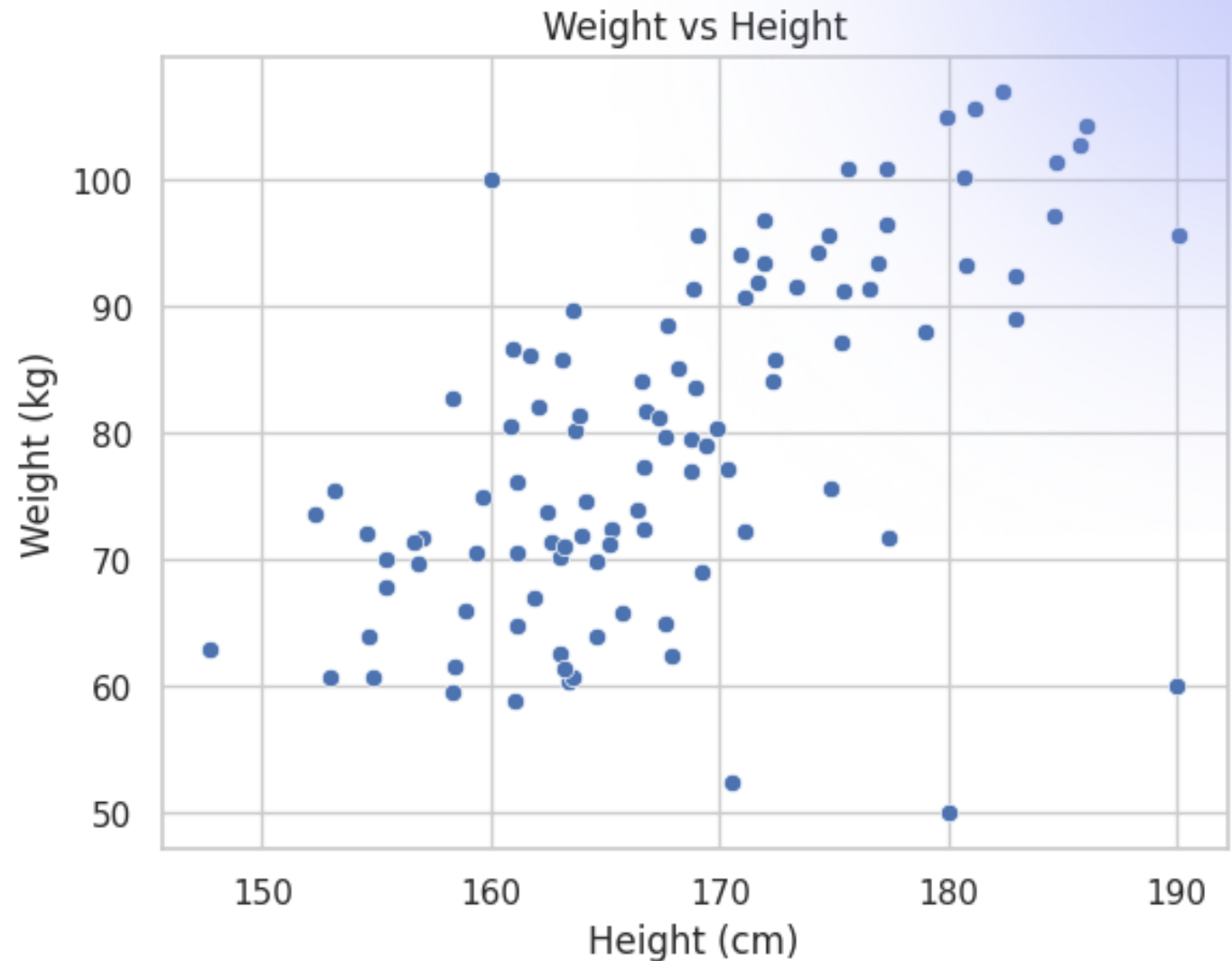


Bivariate analysis

Numeric vs numeric

Scatter Plot

- The standard way to visualize the relationship between two numeric variables.
- Each point represents a single observation.
- Used to identify outliers.
- Effectively highlight complex relationships.

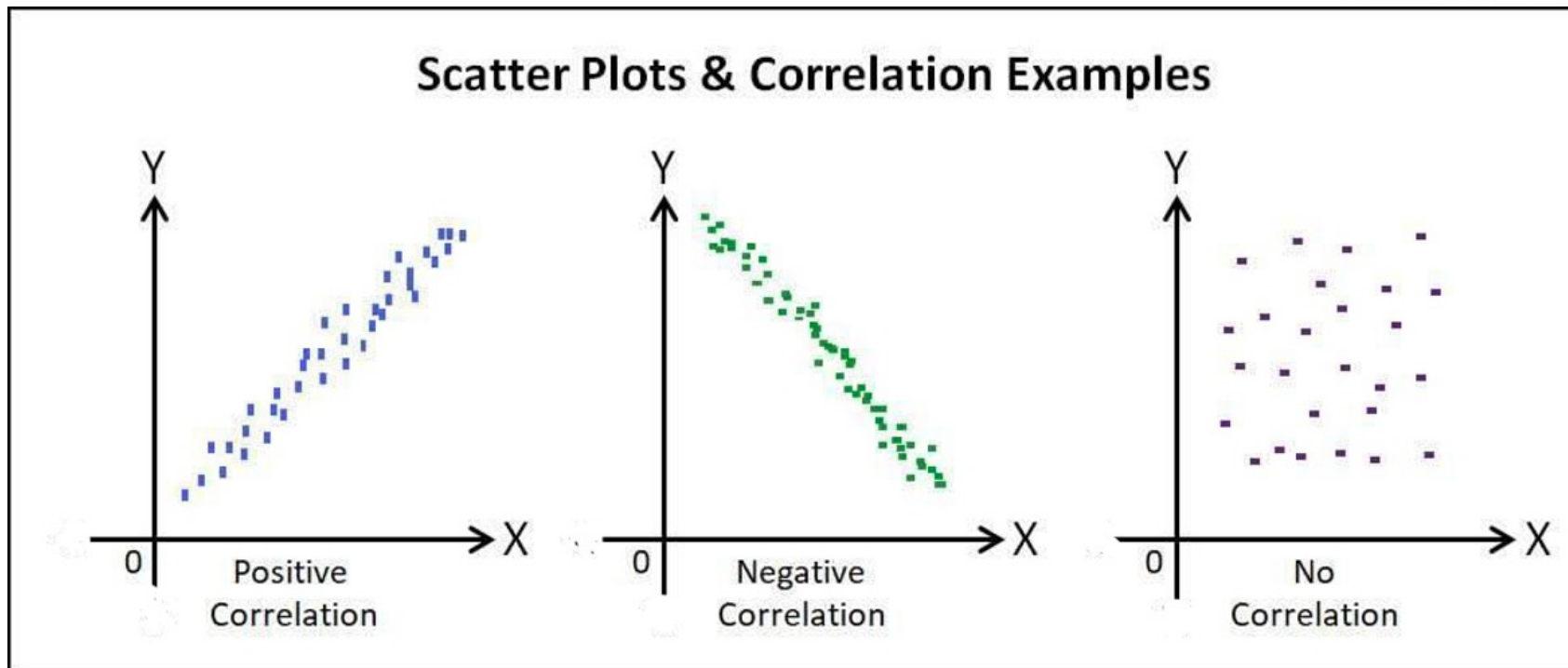


Correlation

Correlation describes the degree to which two variables move in relation to each other.

Variables X and Y are said to be **positively correlated** if high values of X go with high values of Y, and low values of X go with low values of Y.

If high values of X go with low values of Y, and vice versa, the variables are **negatively correlated**.



Pearson's correlation

Pearson's correlation coefficient

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where,

r = Pearson Correlation Coefficient

x_i = x variable samples

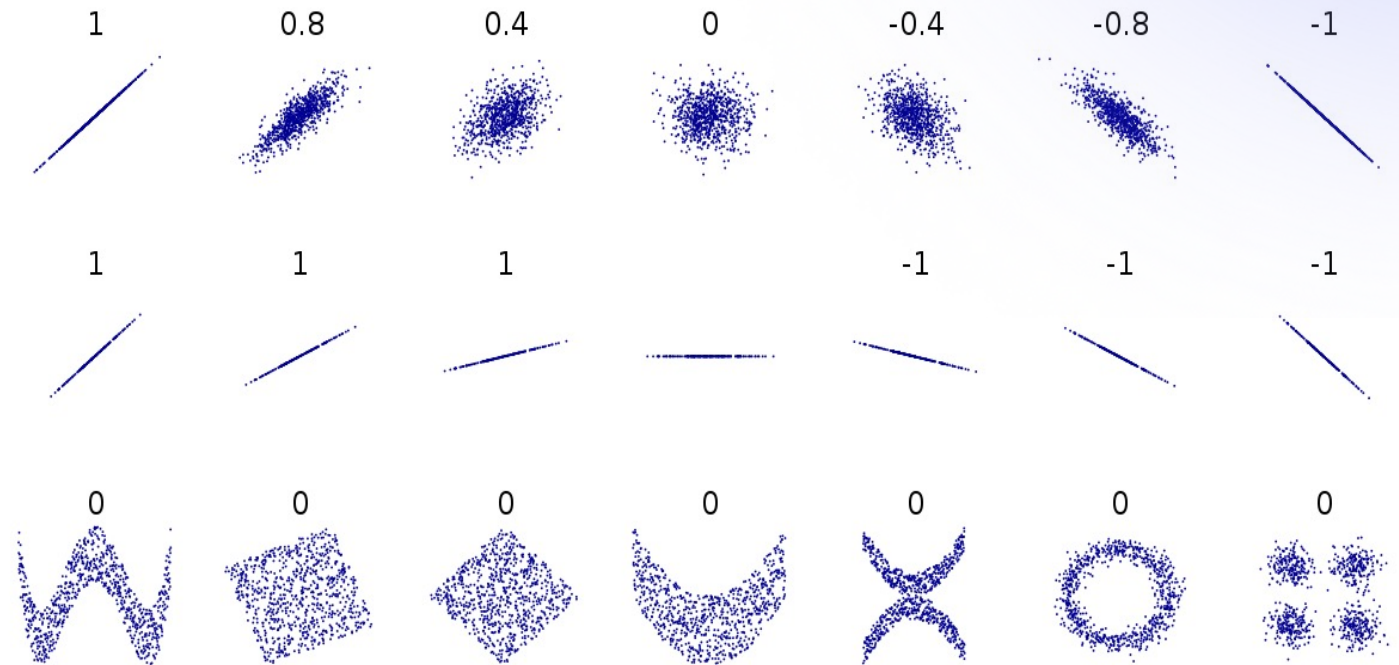
y_i = y variable sample

$\bar{x}$ = mean of values in x variable

$\bar{y}$ = mean of values in y variable

- Measures **linear** relationship between two numeric variables.
- Range: [-1,1]
 $r=1$ perfect positive linear correlation
 $r=-1$ perfect negative linear correlation
 $r=0$ no linear correlation
- Sensitive to outliers

Scatter Plots



Rank correlations

Spearman's Rank Correlation

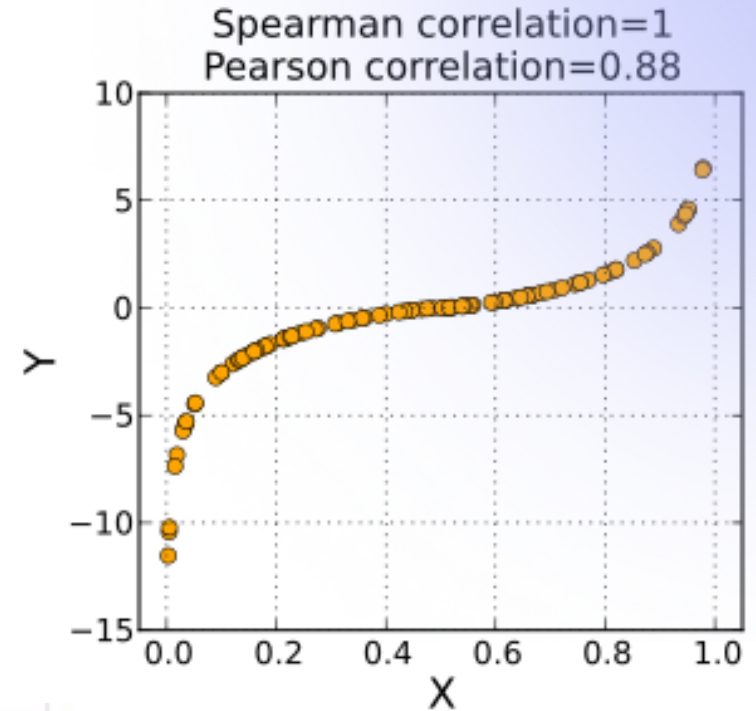
- Describes **monotonic** relationship between numeric variables or categorical ordinal variables.
- Based on the ranked values for each variable.
- Range: [-1,1]
- Robust to outliers

Spearman's Rank Correlation Coefficient

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

d_i – difference between the two ranks of each observation
 n - number of observations

Students	Maths	Science	Students	Maths Rank	Science Rank	d	d square		
A	35	24	A	35	3	24	5	2	4
B	20	35	B	20	5	35	4	1	1
C	49	39	C	49	1	39	3	2	4
D	44	48	D	44	2	48	1	1	1
E	30	45	E	30	4	45	2	2	4
									14

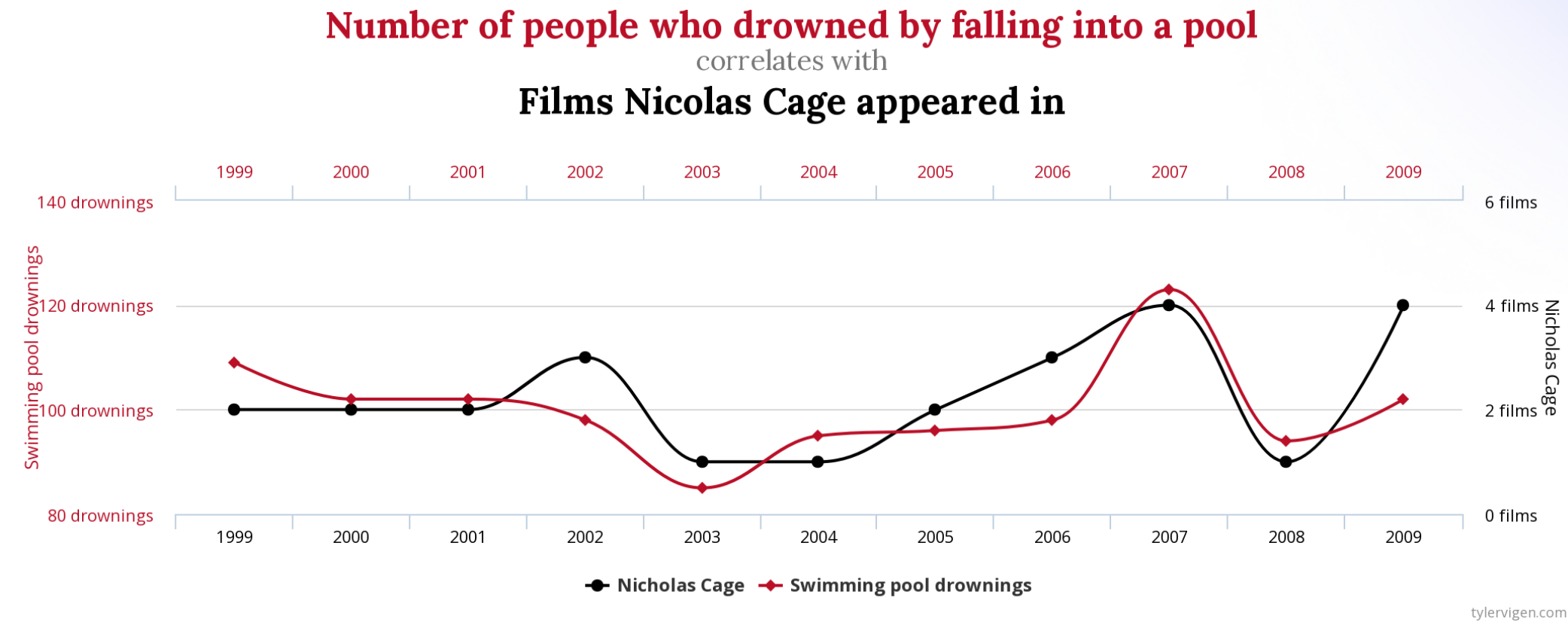


$$\rho = 1 - 6 * 14 / (5 * 24) = 0.3$$

There are other types of correlation, e.g.
Kendall's Rank Correlation

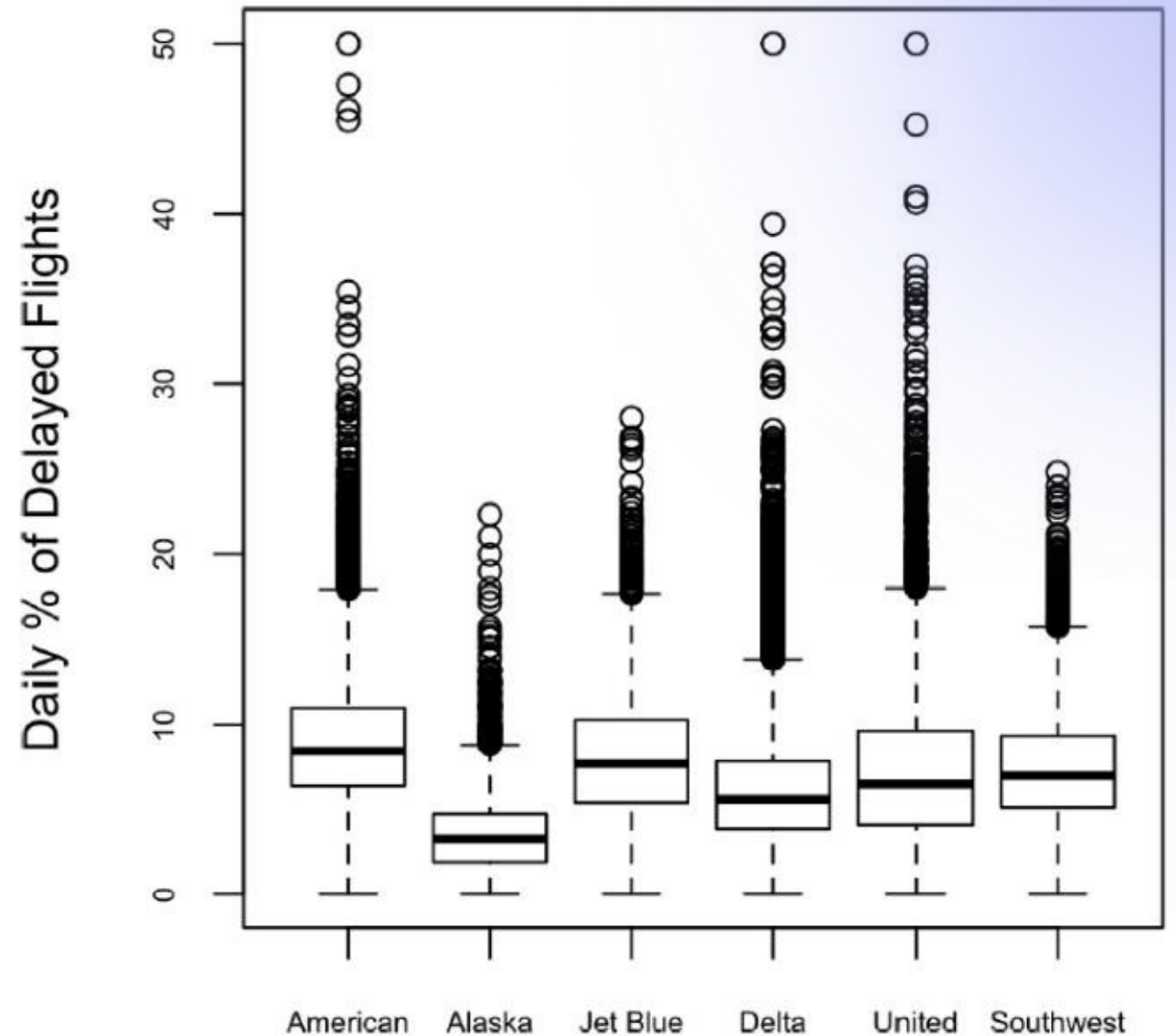
Correlation doesn't mean causation!

Spurious correlations



Numeric vs categorical

Boxplots for visualizing a numeric variable grouped according to a categorical variable

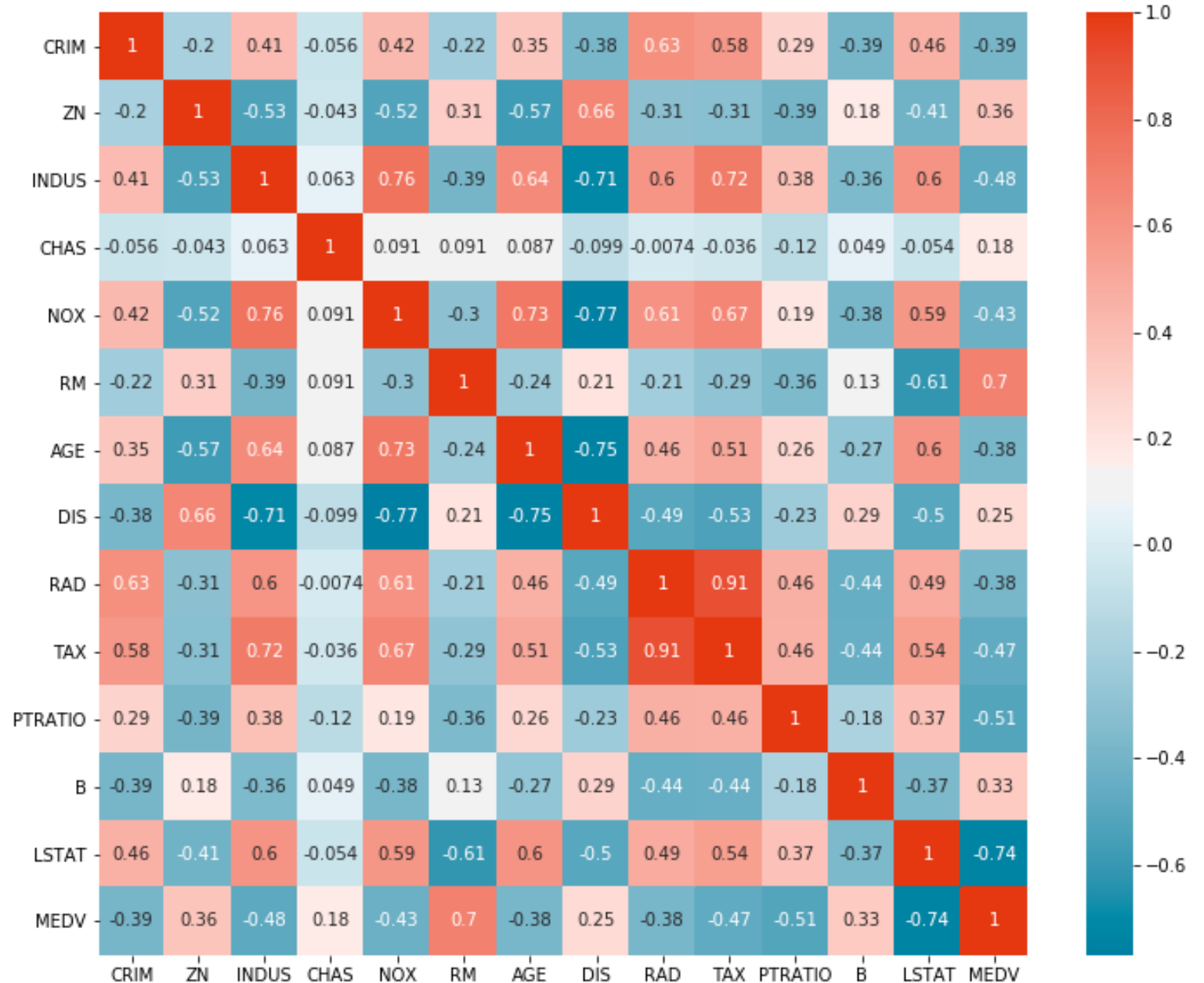


Multivariate analysis

Correlation matrix

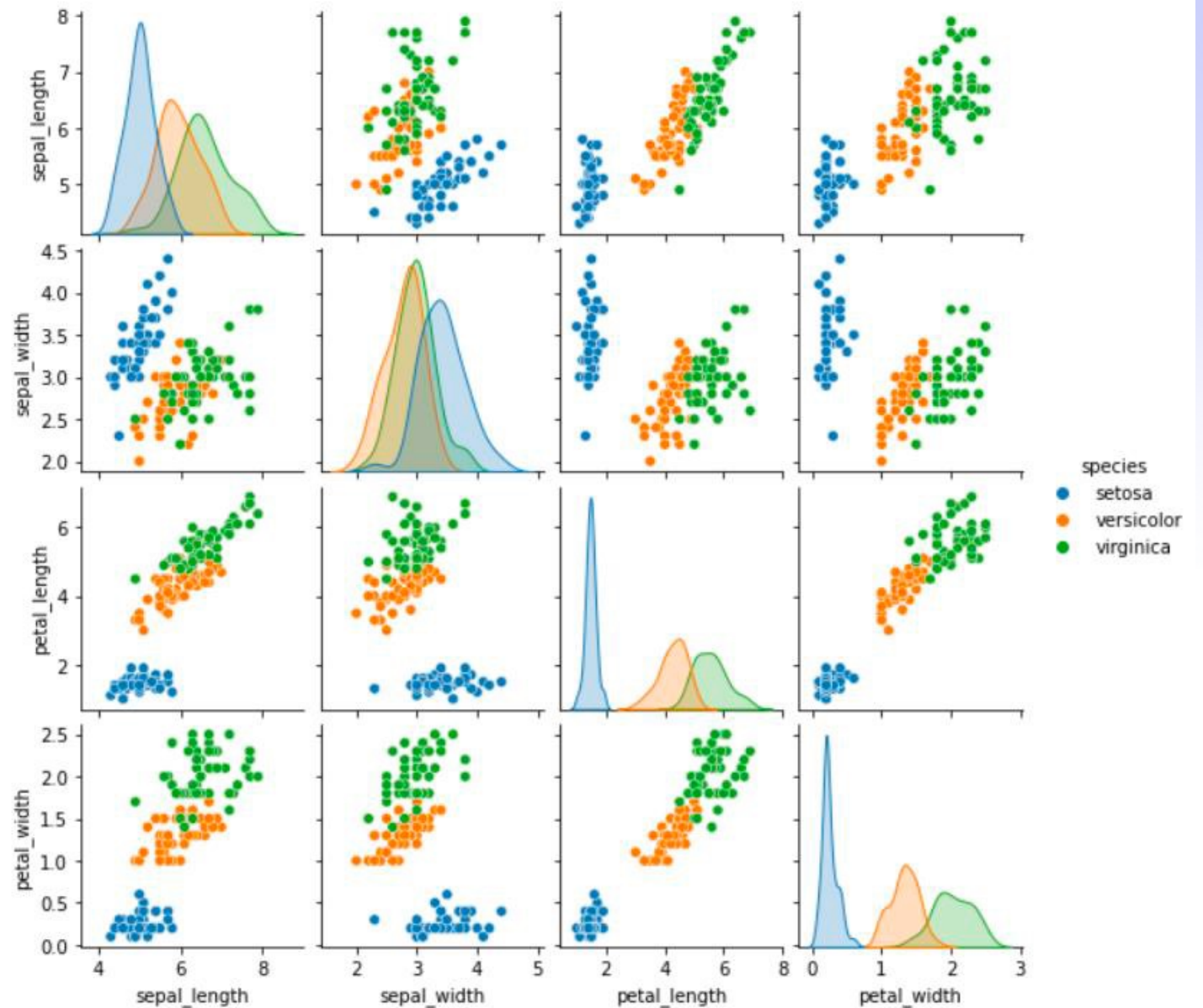
Heatmap

- Multicollinearity- high correlation between independent variables
- For clear visibility of patterns in data use a colour map for the heatmap that represents strong correlations with vibrant colours and weaker correlations near 0 with neutral colours.



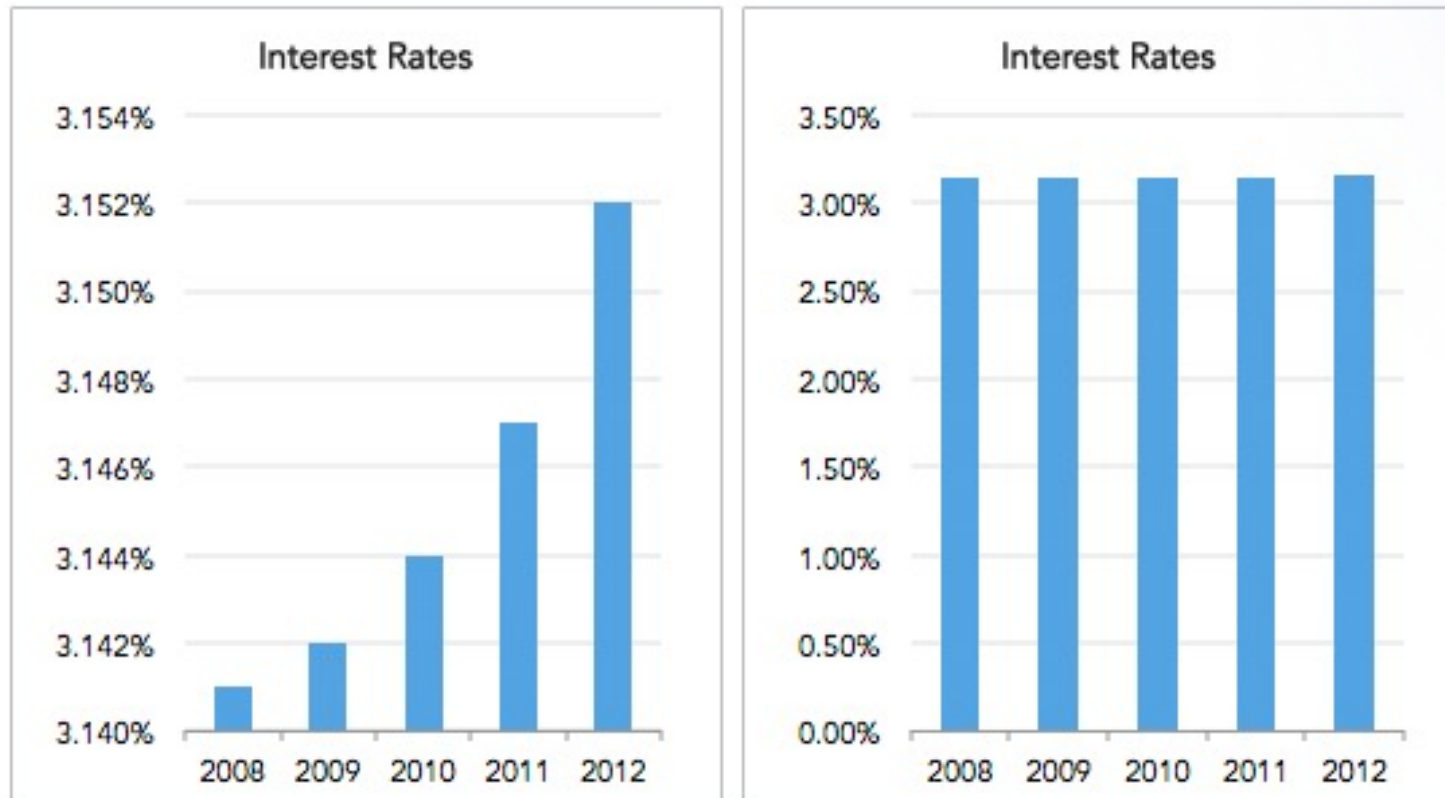
Pair plot

Scatter plot matrix



Axis

Same Data, Different Y-Axis



[The Python Graph Gallery](#) : collection of charts made with Python

Data preparation

- Data Quality
- Untidy Data
- Missing Data
- Encoding
- Scaling

Data Quality

- Validity: data should conform to the defined structure, format and constraints, meet business needs etc.
- Accuracy: data should closely match the real-world information it represents.
- Completeness: data should contain all the necessary information without missing values.
- Consistency: data should be internally consistent, with no contradictions or conflicting information within the dataset.
- Uniformity: data measures should use the same units of measure.

[Wiki page](#)

Untidy Data

- Duplicates
- Wrong data types
- Typos
- Missing data

Duplicates

	ln	dob	gn	fn	is_duplicate
0	SMITH JR	01/03/68	F	WILLIAM	0
1	ROTHMEYER JR	01/03/68	F	WILLIAM	0
2	ASBY JR	01/03/68	F	WILLIAM	0
3	SALTER JR	01/03/68	F	WILLIAM	0
4	SALTER JR	01/03/68	F	WILLIAM	1

Wrong data types

- Zipcode as numeric
- Length as string
- Date as string
- Eye colour as boolean

Type errors and irregularities

- m, Male, man `data['gender'] = data['gender'].replace(['Male', 'man'], 'm')`
- data science, Data Science, DATA SCIENCE
`string.capitalize()`
`string.upper()`
`string.lower()`
- (+34) 459-66-666, (0034) 459-66-666, 003445966666
- "Toyota Corolla",
" Toyota Corolla "
- 000163647, 163647
`string.strip()`

Fuzzy matching

Record #	Name
A	Hewlett Packard
B	HP, Inc.
C	Hewlett-Packard Corp
D	HP Corporation
E	Hewlet Pakkard
F	Hewwlet Packerd



Hewlett Packard

Fuzzy string matching is the technique of finding strings that match a pattern approximately (rather than exactly)

Fuzzy string matching with python

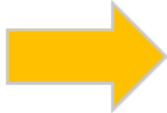
Missing data

Approaches to deal with missing data:

1. Remove entire predictor(s) and/or sample(s) that contain missing values:
 - Drop column
 - Drop row
2. Imputation. Use information in the dataset to estimate the values of other predictors.
 - Estimates (e.g. mean, median, mode)
 - Models (KNN-imputation)
 - Time Series fill
3. Mark with special symbol (“unknown”)

Encoding

Label Encoding



color	color
red	0
green	1
blue	2
red	0

Pros:

- Simplicity
- Efficiency

Cons:

- Ordinal assumption
- ML algorithm misinterpretation

Ordinal Encoding

```
desired_order = ['High School', 'Bachelor', 'Master', 'PHD']
```

	education	education_encoded
0	Bachelor	1
1	High School	0
2	Master	2
3	Bachelor	1
4	High School	0
5	PHD	3
6	Master	2

Pros:

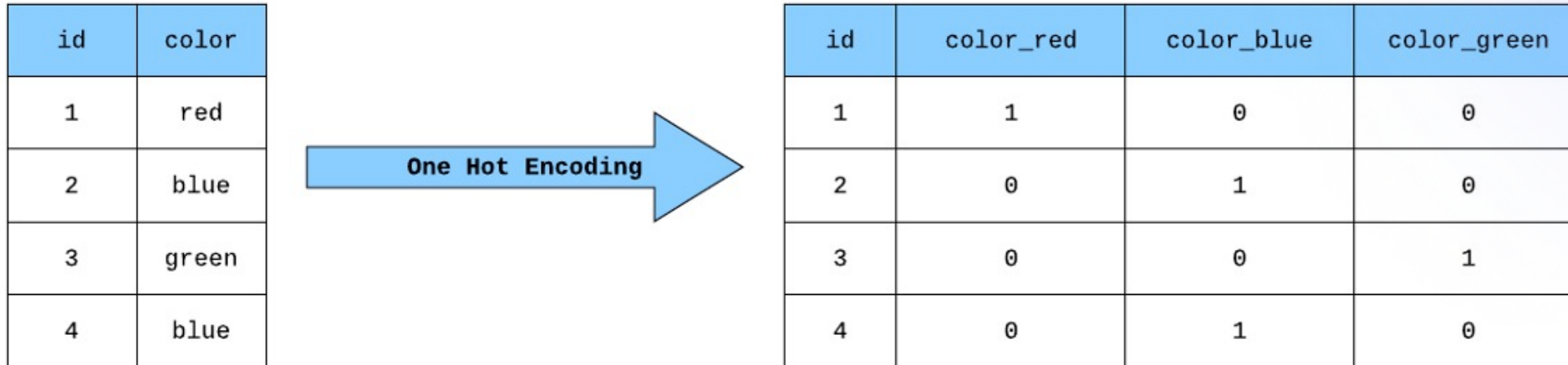
- Maintains ordinality
- Efficiency

Cons:

- Manual mapping
- ML algorithm misinterpretation

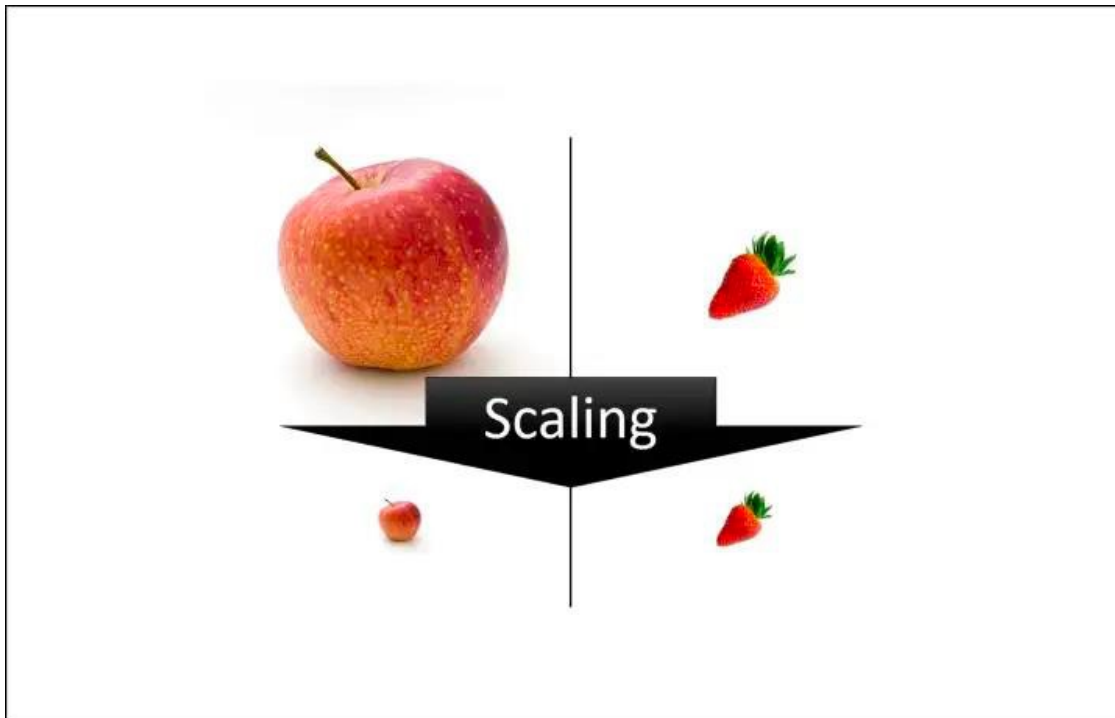
Encoding

One-Hot Encoding

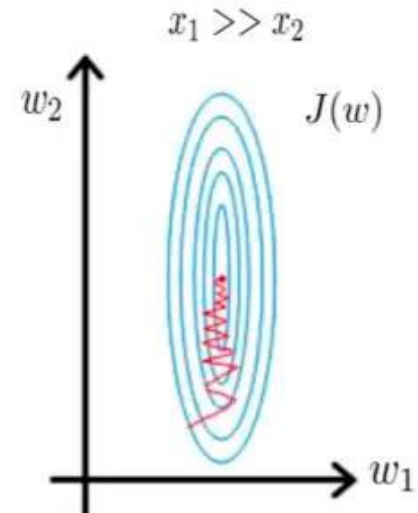


The ***Dummy Variable Trap*** - one input variable can be perfectly predicted by others (multicollinearity). To avoid it, use one less dummy variable than the number of categories.

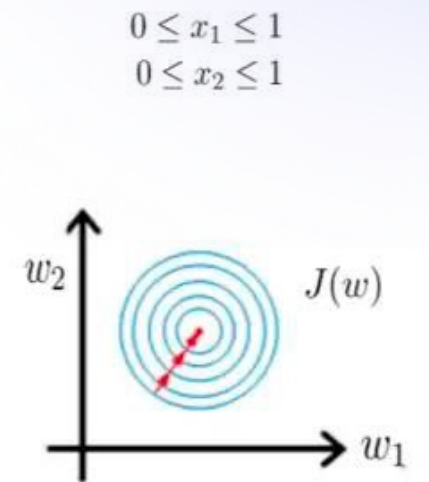
Scaling



Gradient descent
without scaling



Gradient descent
after scaling variables



Scaling

Min Max scale

$$x_{new} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Standard scale

$$x_{new} = \frac{x - \mu}{\sigma}$$

Vector length as scale

$$x' = \frac{x}{\|x\|}$$

Normalization vs Standardization

Normalization - rescaling the values into a certain range.

Useful for algorithms that weight inputs like Neural Networks and algorithms that use distance measures (e.g. K-Nearest Neighbors (KNN)).

Standardization - rescaling data to have a mean of 0 and std of 1.

Useful for algorithms that assume data has a Gaussian distribution and that all features have the same variance (e.g. Support Vector Machines (SVMs), Linear Regression).

Feature Selection

Filter (EDA) Methods

- Useless (unique data)
- Constant
- Quasi constant
- Correlations

Wrapper methods

- treat feature selection as a search problem, evaluating subsets of features based on the model's performance

Embedded methods

- integrate feature selection into the model-building process itself, optimizing during training

Feature Engineering

Feature Engineering is the process of creating new features or modifying existing features to improve the performance of machine learning algorithms.

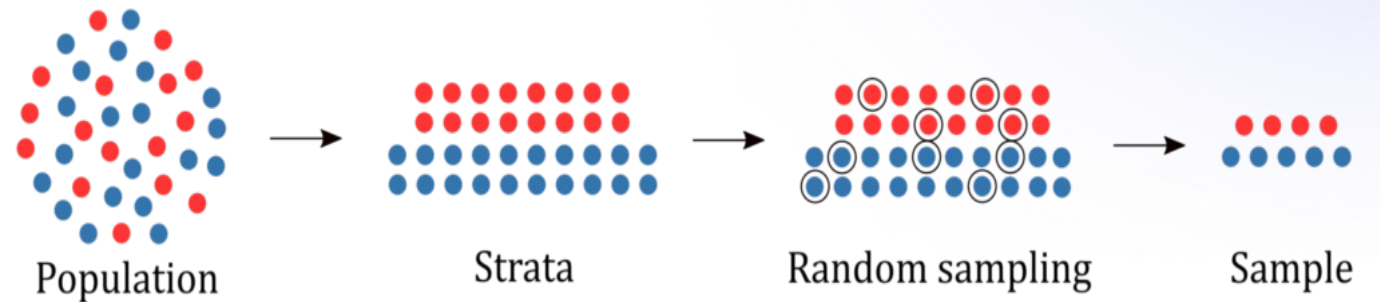
- Feature generation (e.g. rent/per_square_meter, numerics to categories ($\geq 5K$, $< 5K$))
- Feature transformation (e.g. square, log, square root)

Sampling methods

Simple random sampling



Stratified sampling



Other sampling techniques

Dimensionality reduction: simplifying complexity

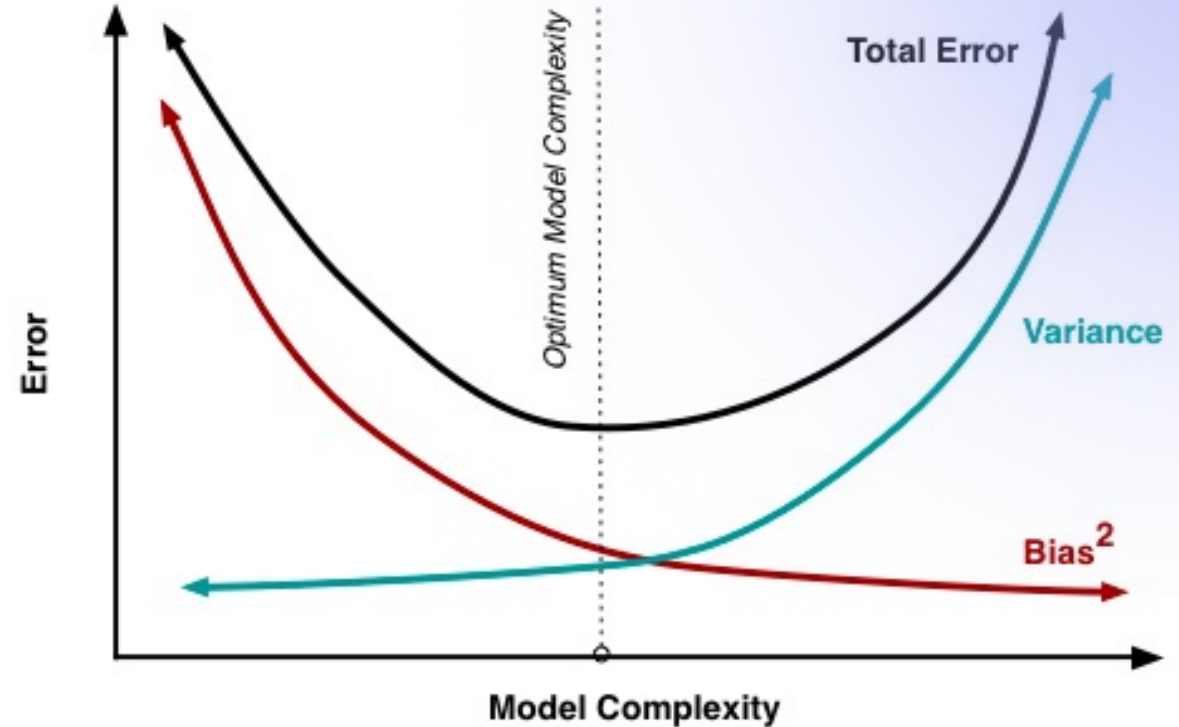
Dimensionality reduction is concerned with reducing the number of input features.

Reasons

- Time and required storage space reduction
- Model complexity
- Visualization

Large numbers of input features can cause poor performance for machine learning algorithms.

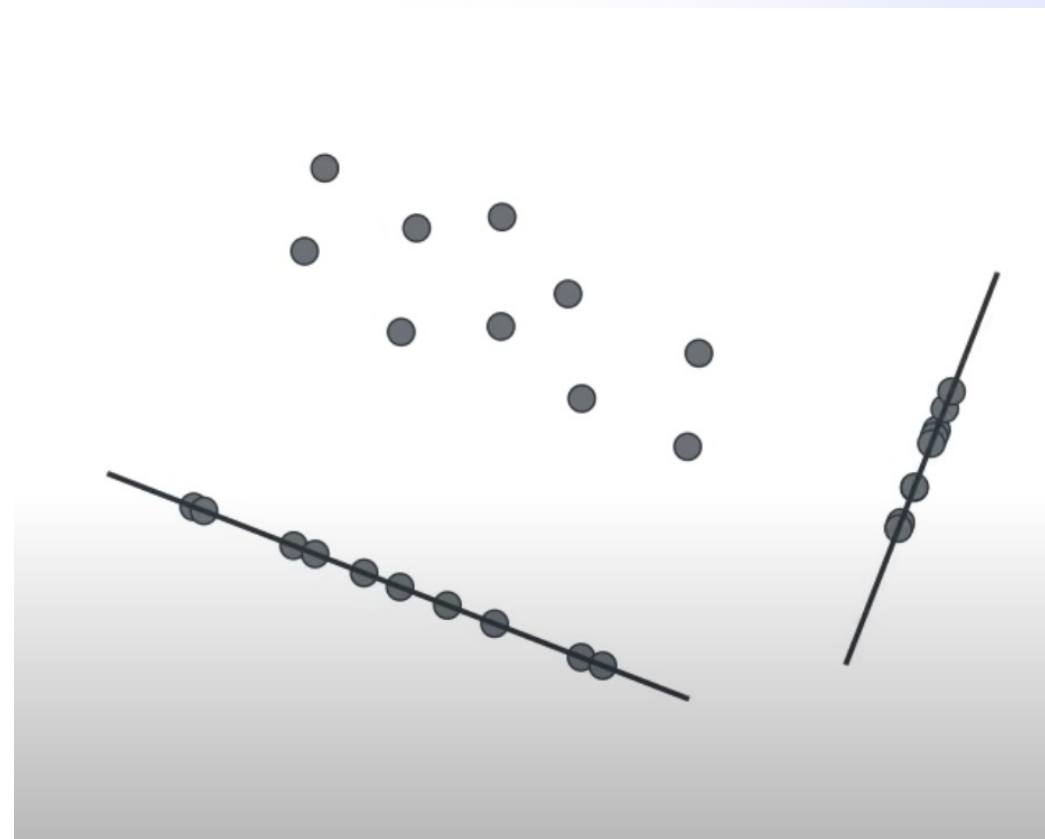
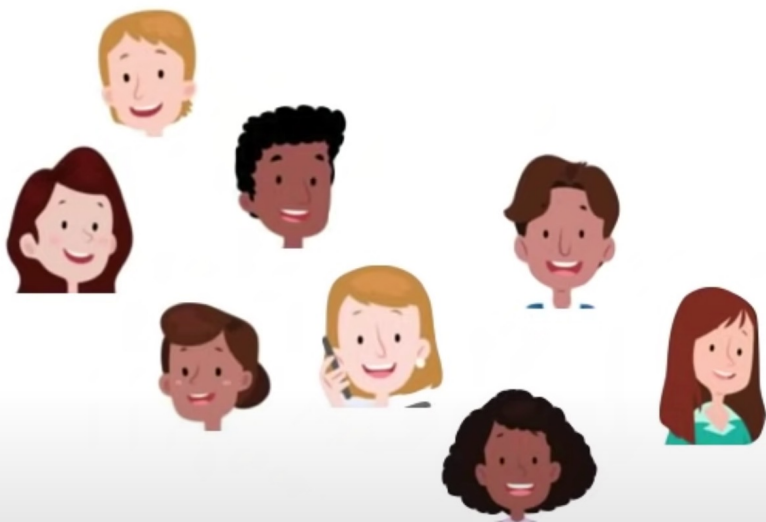
Dimensionality reduction methods include feature selection, linear algebra methods, projection methods and autoencoders



PCA
t-SNE

Principal Component Analysis (PCA)

Taking a picture



PCA

- linear transformation
- finds PC's (principal components) - top k orthogonal directions that explain the most variance in the data
- PC's are eigenvectors (sorted in descending order according to their corresponding eigenvalues) of covariance matrix
- standardize data before PCA

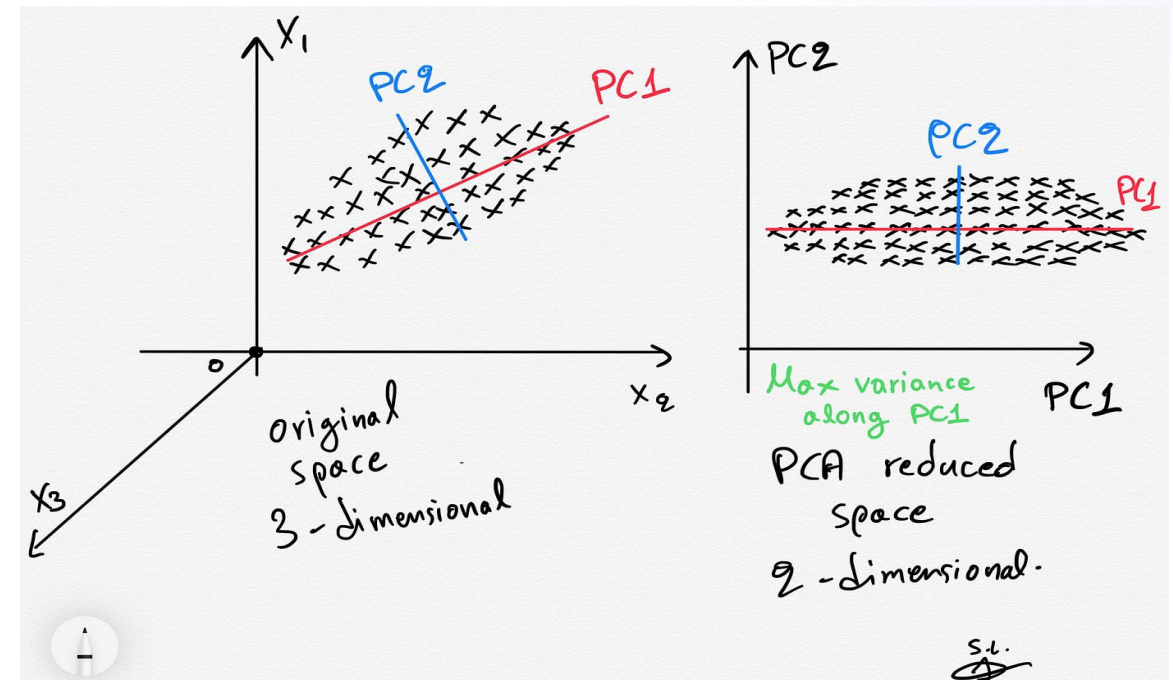
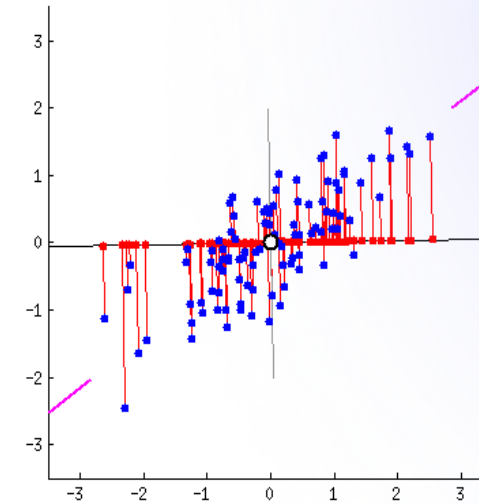
$$\text{cov}(X, Y) = \frac{1}{n} \sum_{t=1}^n (X_t - \bar{X})(Y_t - \bar{Y})$$

Covariance matrix

$$\Sigma = \begin{pmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{pmatrix}$$

PCA explained

<epam>

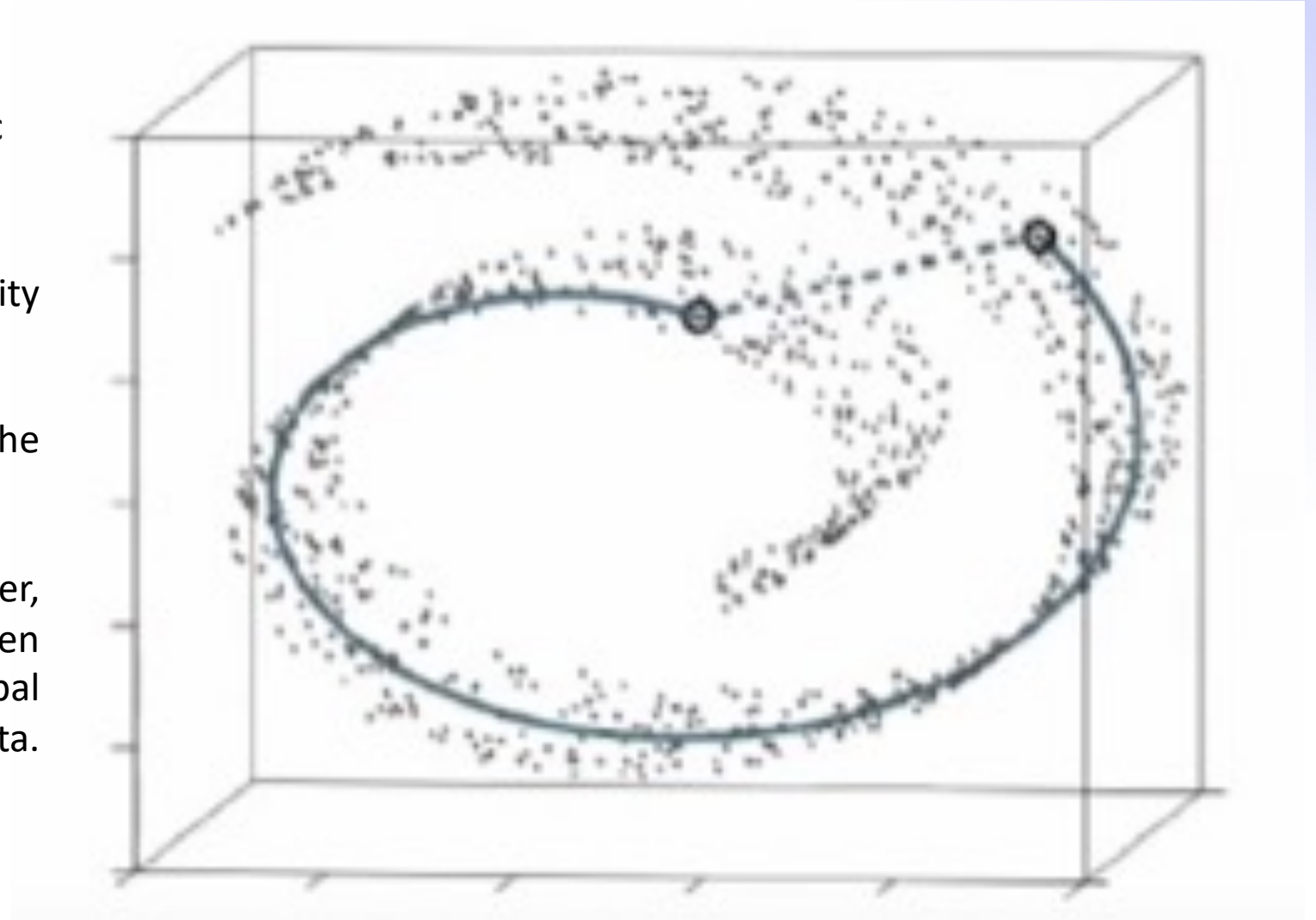


Linear vs Nonlinear dimensionality reduction

t-SNE (t-Distributed Stochastic Neighbor Embedding)

- Non-linear dimensionality reduction technique
- Can preserve local structure of the data
- Perplexity - key hyperparameter, determines the balance between preserving the local and global structure of the data.

t-SNE explained





Thank you for your attention!

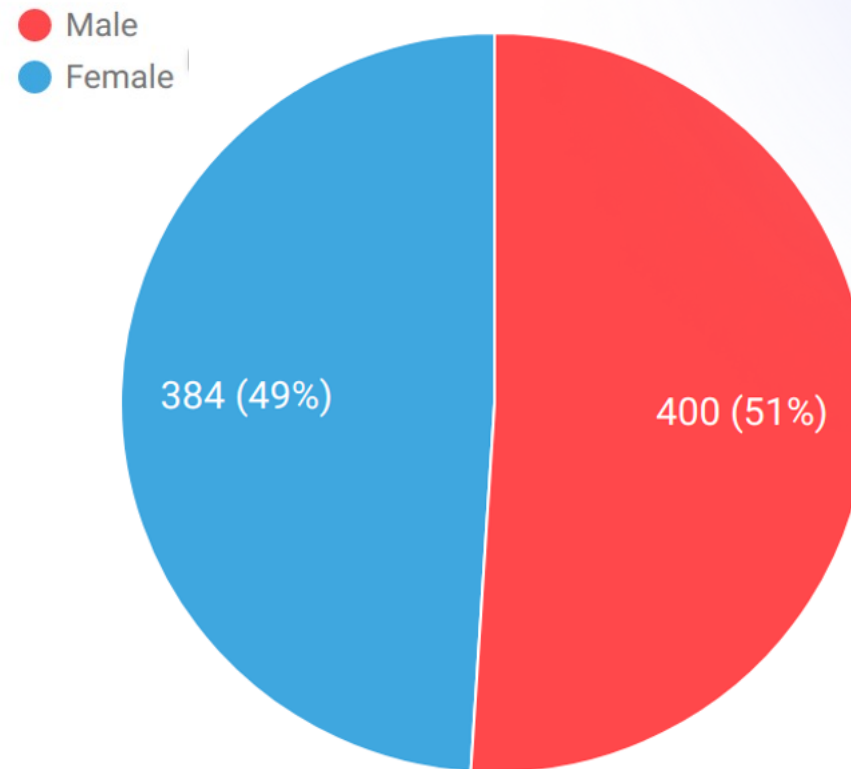
Iya Ivanishko

DS basics, 2023

Appendix

Pie chart

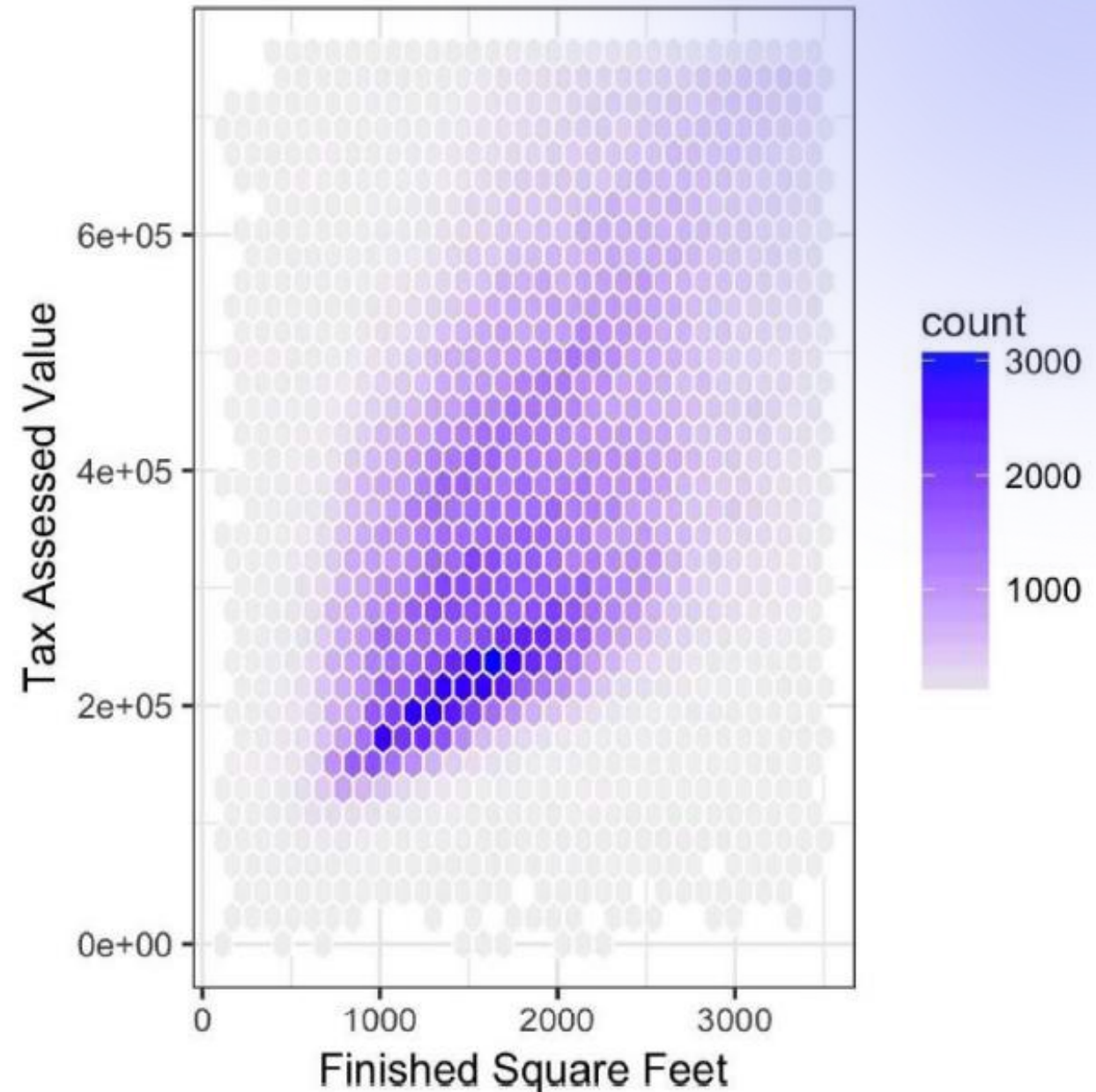
- Best used when the number of categories is relatively small.
- Gives a quick visual impression of relative proportions.



Hexagonal Binning

Scatterplots are fine when there is a relatively small number of data values

For data sets with hundreds of thousands or millions of records, a scatterplot will be too dense

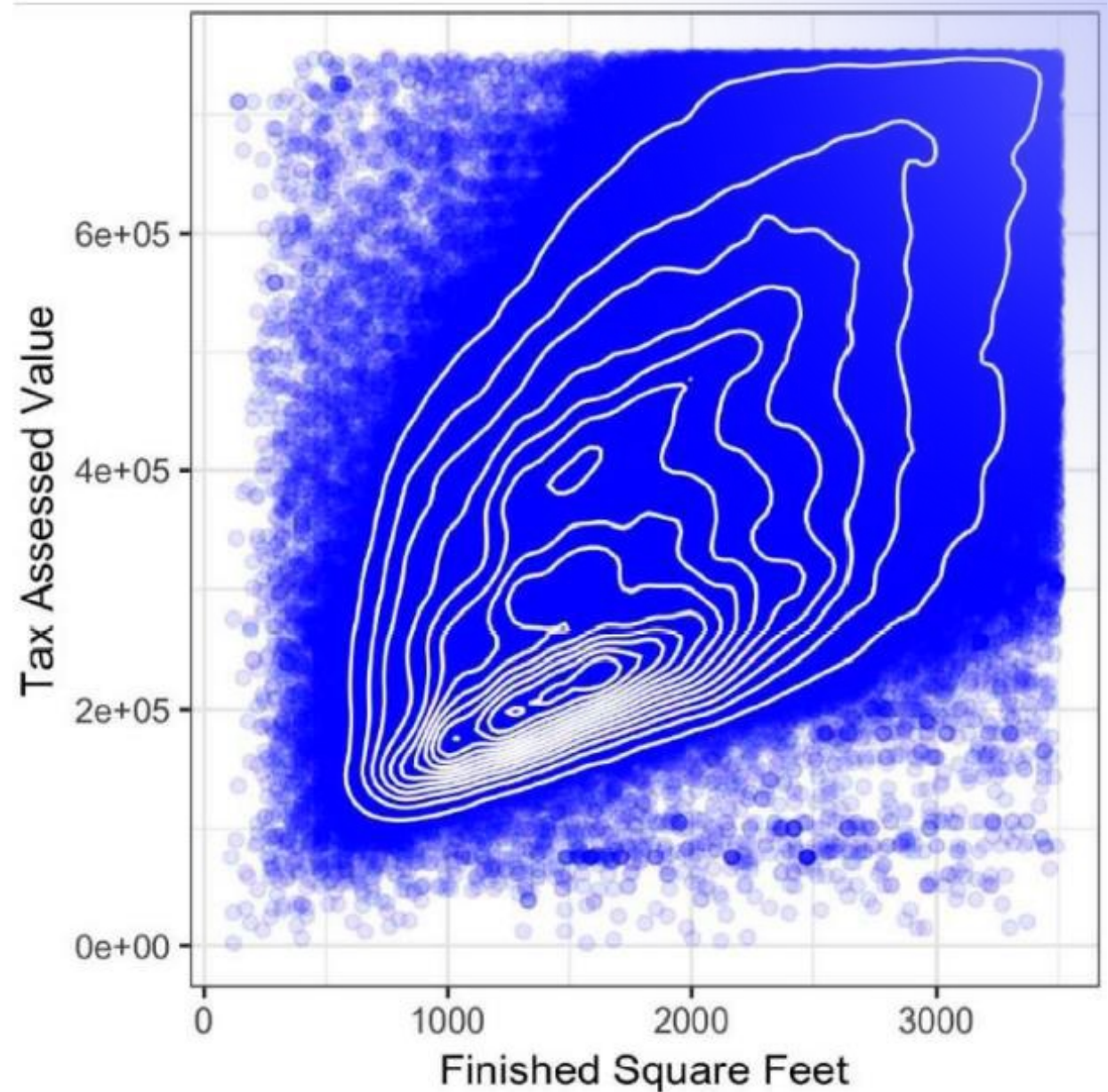


Contour Plot

The contours are essentially a topographical map to two variables

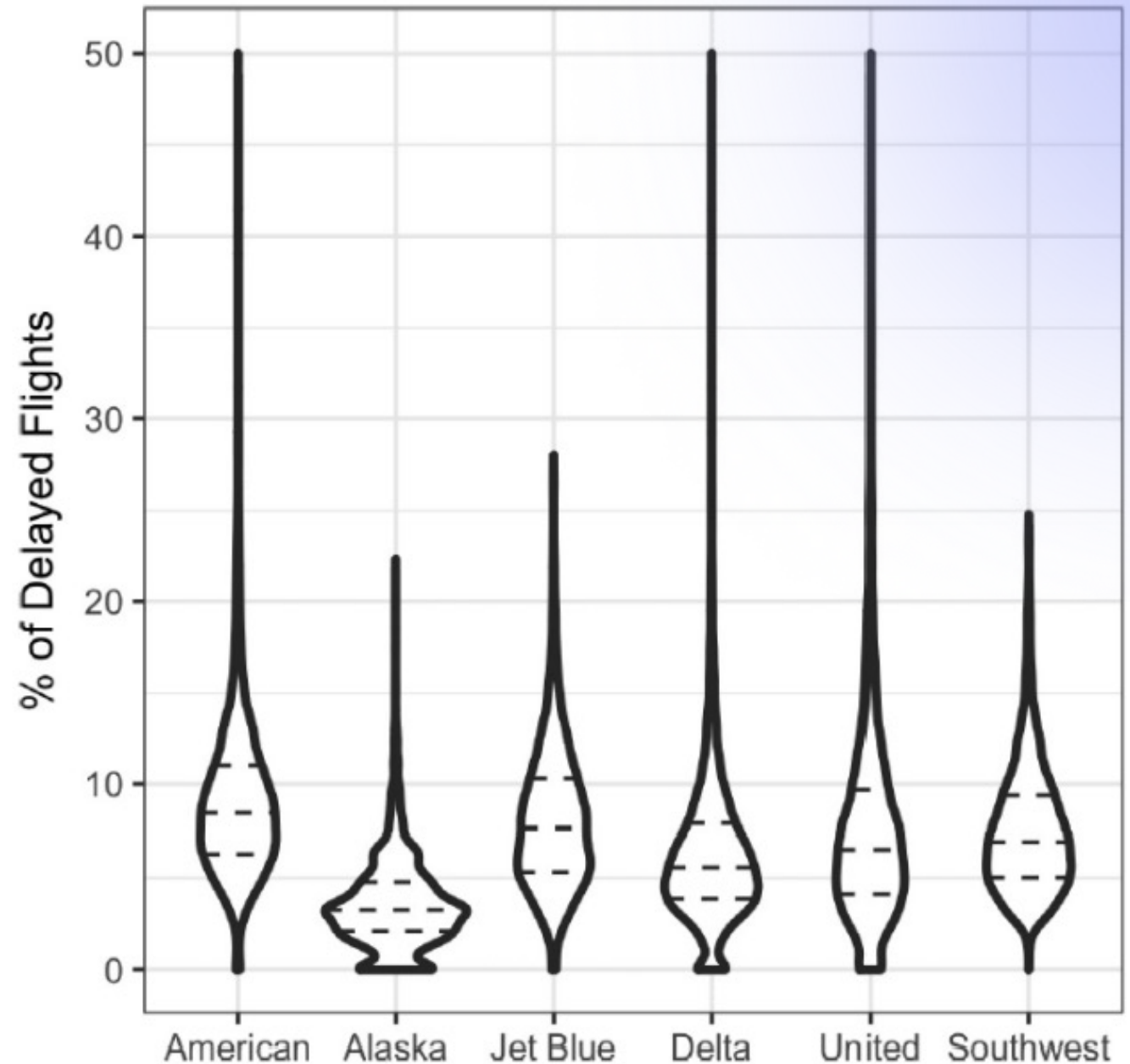
- each contour band represents a specific density of points, increasing as one nears a “peak.”

This plot shows a similar story as previous, there is a secondary peak “north” of the main peak.



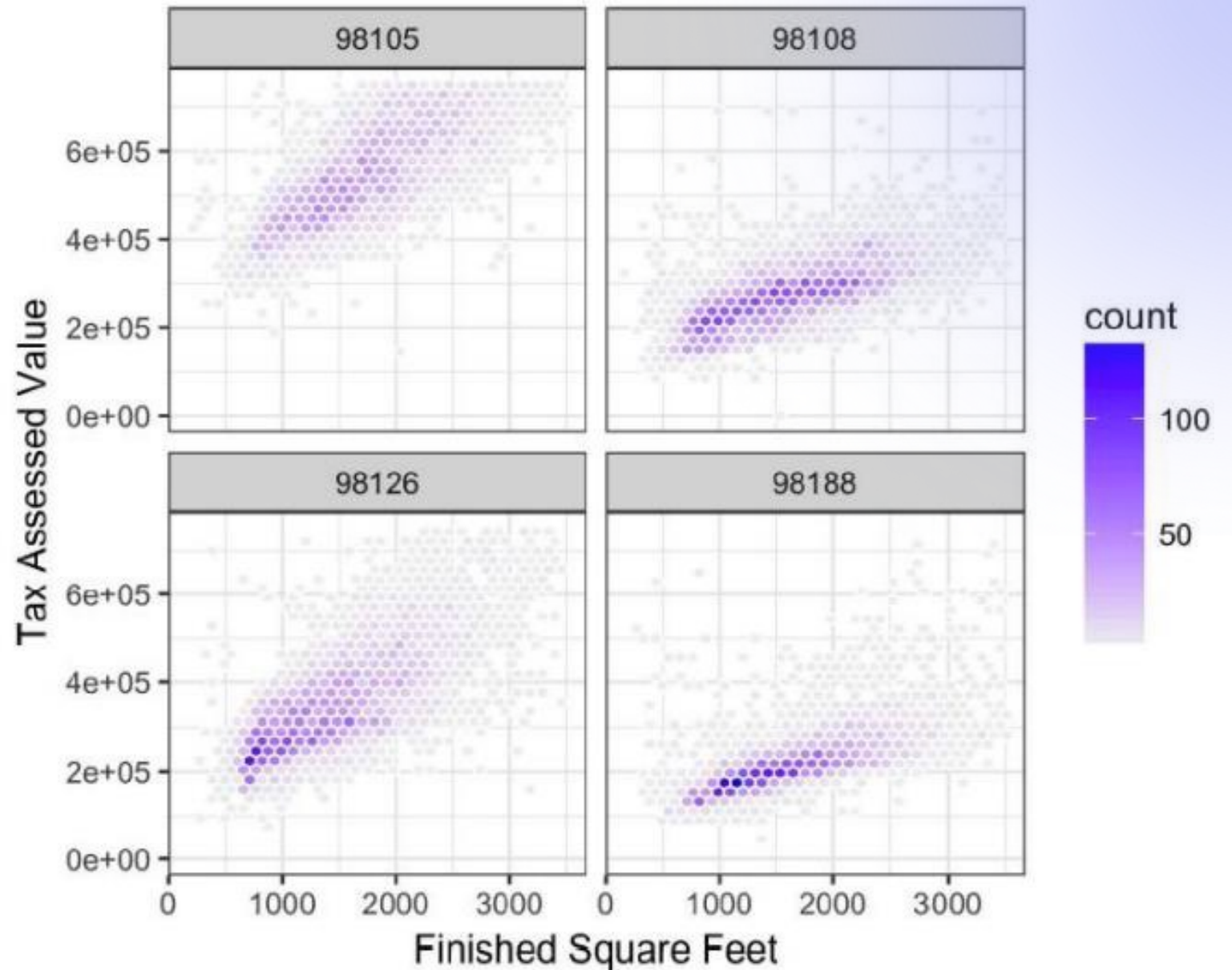
Violin Plot

A violin plot, is an enhancement to the boxplot and plots the density estimate with the density on the y-axis.



Using categories as grids

We can use categories, in this example zip code, to make multiple plots



Correlation matrix Heatmap

Misleading colours

